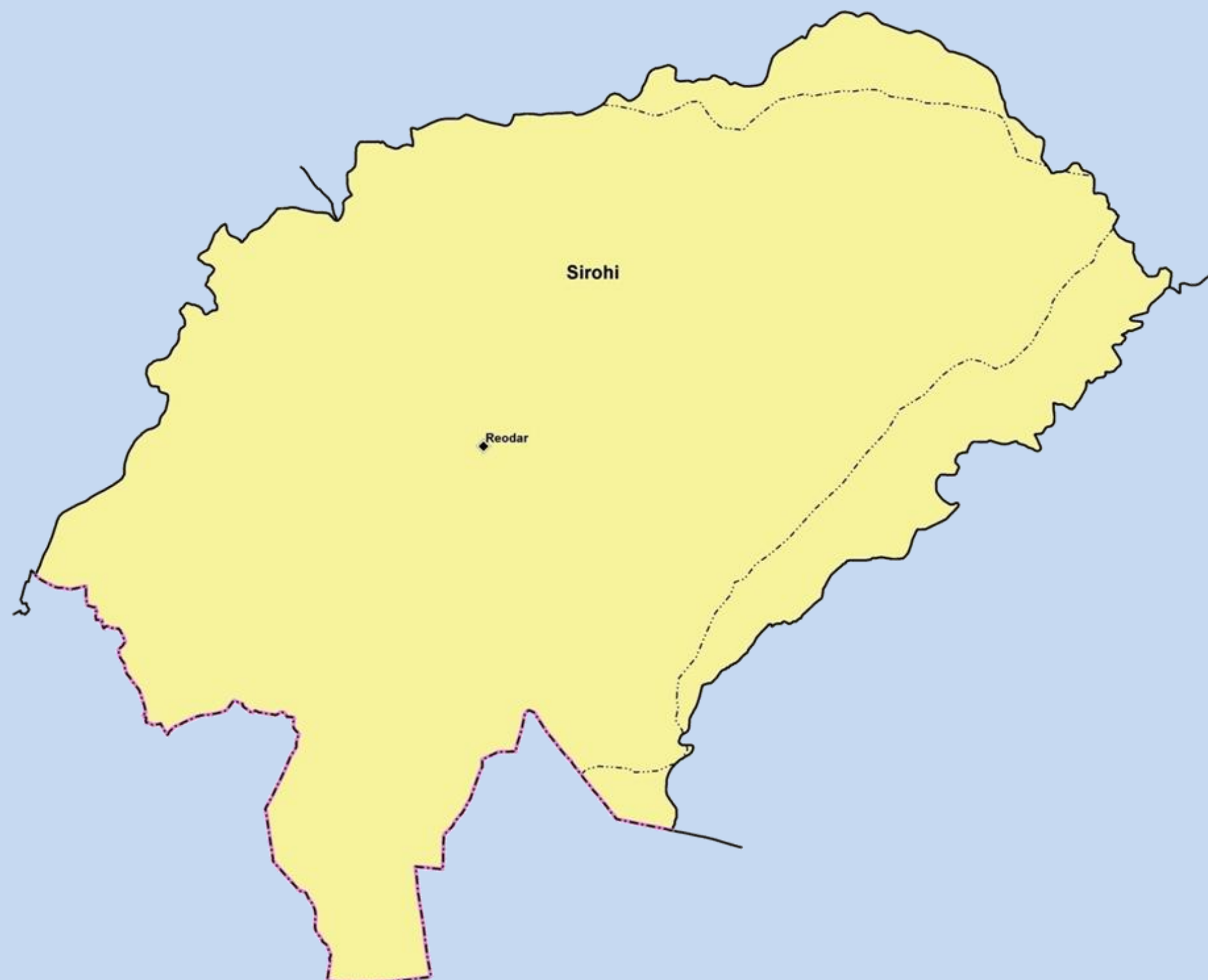


Hydrogeological Atlas of Rajasthan

Sukli River Basin



2013

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

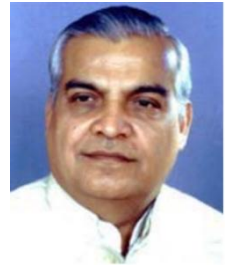
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

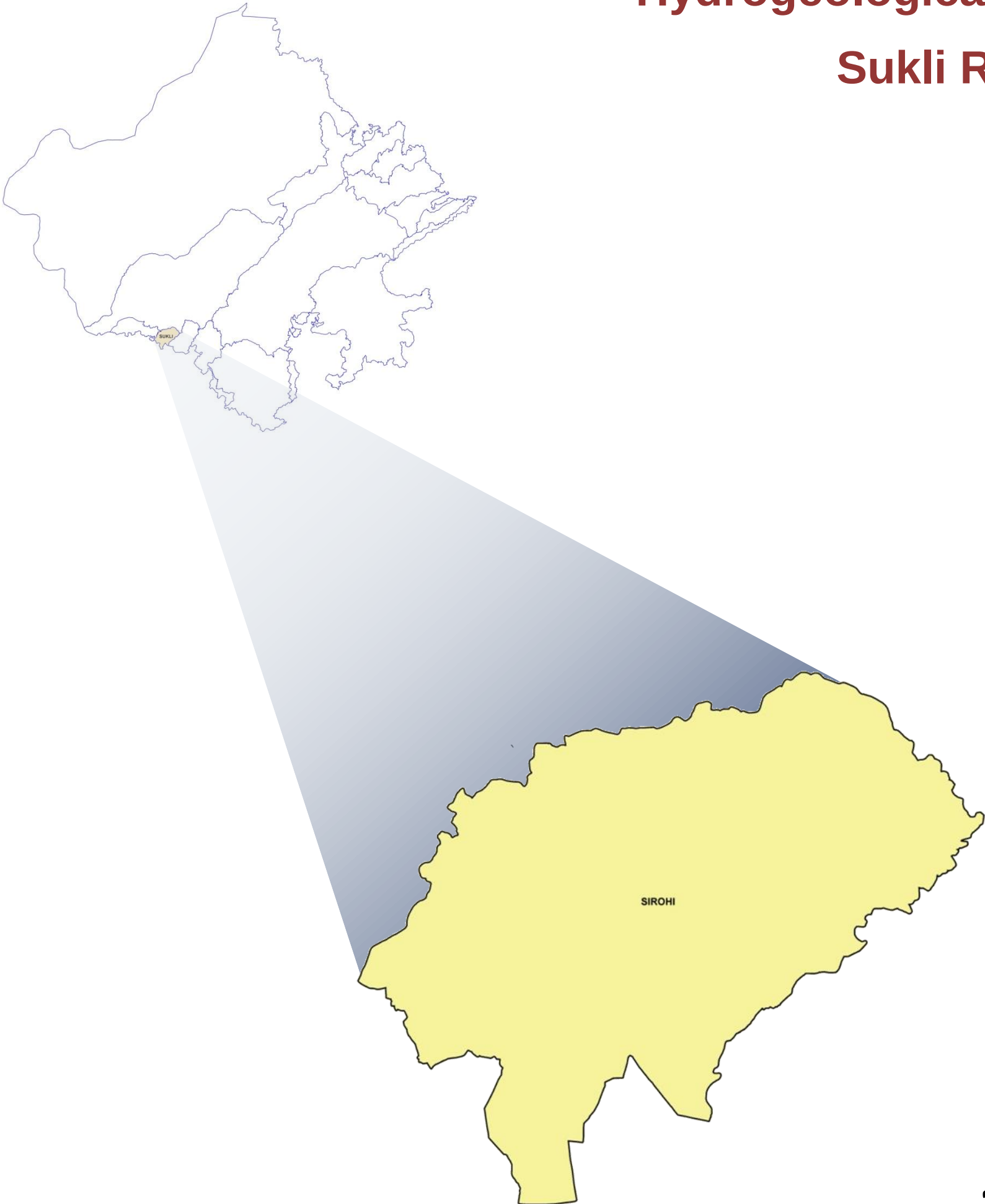
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Sukli River Basin



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ADMINISTRATIVE SETUP

SUKLI RIVER BASIN

Location:

River Sukli (also called Sipu River) flows in the south-western part of Rajasthan, and its basin is very small in geographic extend measuring less than 1,000 sq kms. It stretches between 24° 24' 42.01" to 24° 47' 22.30" North latitudes and 72° 19' 43.99" to 72° 48' 32.40" East longitudes (Plate – I). It is bounded on the north by the Luni River Basin, on the east by the West Banas River Basin and on the west by the area drained by Other Nallahs of Jalore. The southern border is shared with Gujarat State. The basin extends over parts of Sirohi district only. The total catchment area of the Basin is 999.4 sq km.

The western part of the basin is marked by hilly terrain belonging to the Aravali chain. East of the hills lies a narrow alluvial plain with a gentle eastward slope. The River Sukli originates in the western slopes of Mt. Abu, the Silari hills and the area between these two hill features. It flows parallel to Mt. Abu and the West Banas River for about 29 km in Rajasthan and joins the West Banas River in Gujarat State.

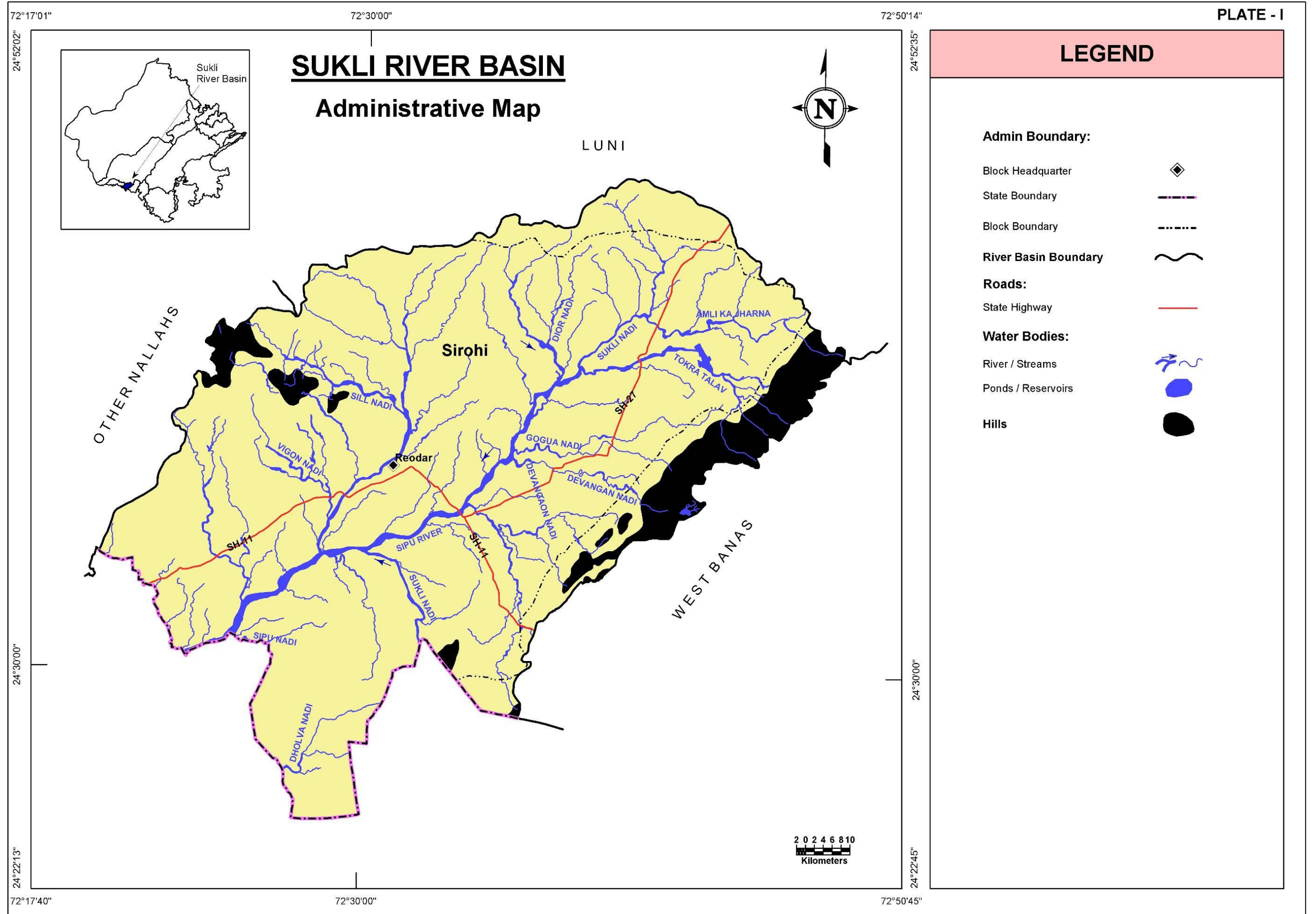
Administrative Set-up:

Administratively, Sukli River Basin extends over parts of Sirohi, encompassing 3 Blocks and 114 villages.

S. No.	District Name	Area (sq km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Sirohi	999.4	100.0	3	114

Climate:

The Sukli River basin is very small and falls at the edge of arid climatic region border. By and large it is a dry area but not much arid since it is surrounded by hills. It is extremely cold from October to February while turning hot from March to September. Summer records very high temperatures in the range of 47°C – 48°C but winters are chilly when temperature falls to about 2°C. The basin receives fairly low rainfall because the mean annual rainfall over Sukly River Basin was 350 mm, of which most of the rainfall is received during the four Monsoon months (June-September).



TOPOGRAPHY

The basin is bordered in the east by Aravalli Range trending northeast – southwest and Mount Abu, the highest point of Rajasthan falls within this basin. There are some east west trending hills in the western part of the basin and some isolated local hills within the basin. While the highest point of the basin is about 1690m above mean sea level, the lowest point is about 192m amsl located at the exit point of the Sukli river from Rajasthan, into Gujarat. The rest of the basin area is undulating with elevations ranging between 200m amsl to 300m amsl. The Sukli river flows in a northeast – southwest direction, while receiving tributaries from east, north and west directions.

Table: District wise minimum and maximum elevation

S. No.	District Name	Min Elevation (m amsl)	Max. Elevation (m amsl)
1	Sirohi	191.7	1,690.2

RAINFALL

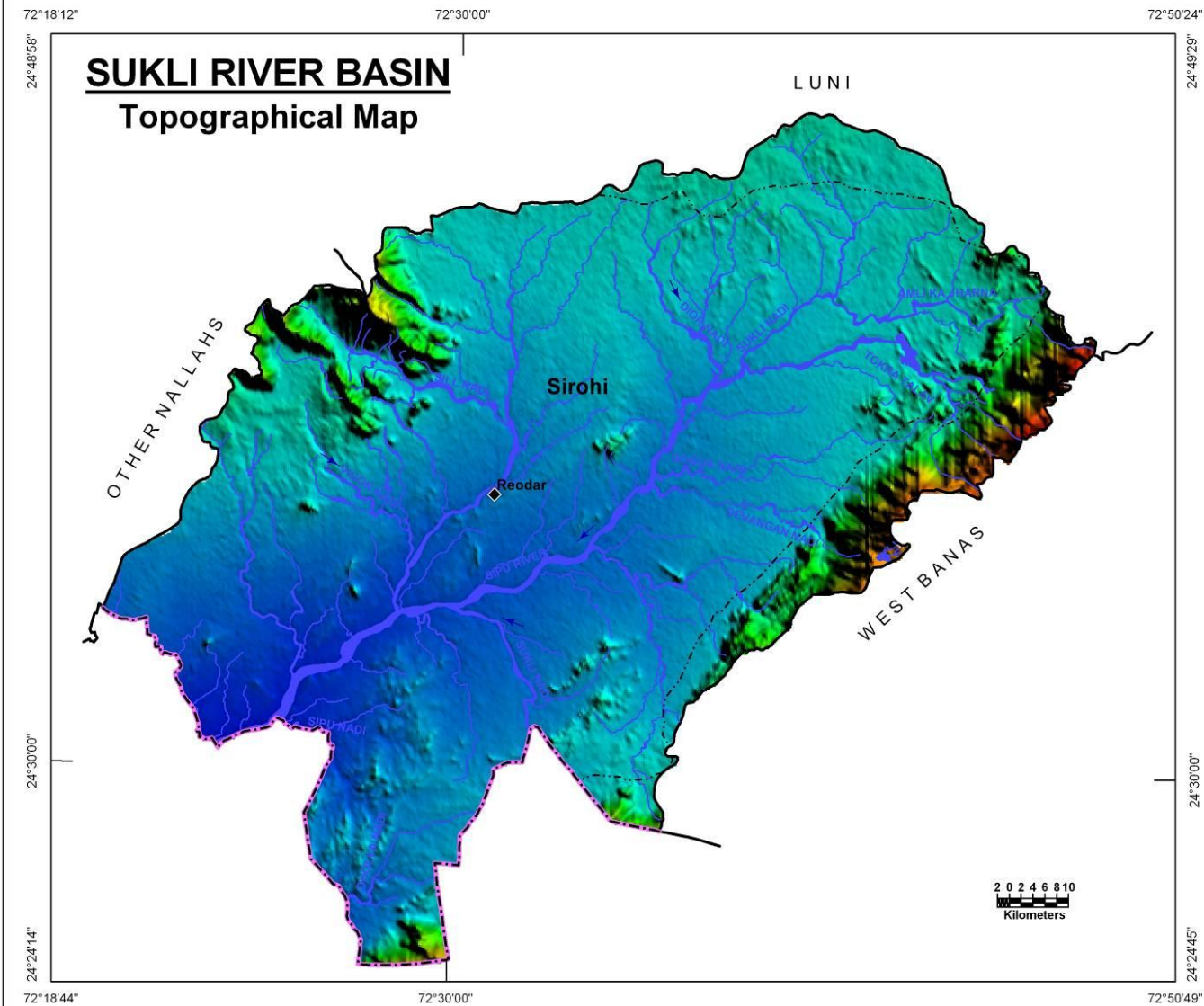
The general distribution of rainfall across the Sukli River Basin (as per rainfall statistics for the year 2010) can be visualized from isohyets presented in the Plate III where most of the northern, eastern and western parts receive higher rainfalls (in the range of >1000mm of total annual rainfall) whereas; the southern parts received relatively low rainfall (between 800 to 1000mm). The average annual rainfall for the basin computed based on available station data is about 1281 mm. The rainfall data for available rain gauge stations is presented below.

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajabsthan.gov.in>))

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
1	Mount Abu	1,484.0	187.8	1,671.8
2	Reodhar	710.0	180.0	890.0



PLATE - II



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

Block Boundary

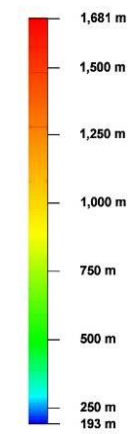
River Basin Boundary

Water Bodies:

River / Streams

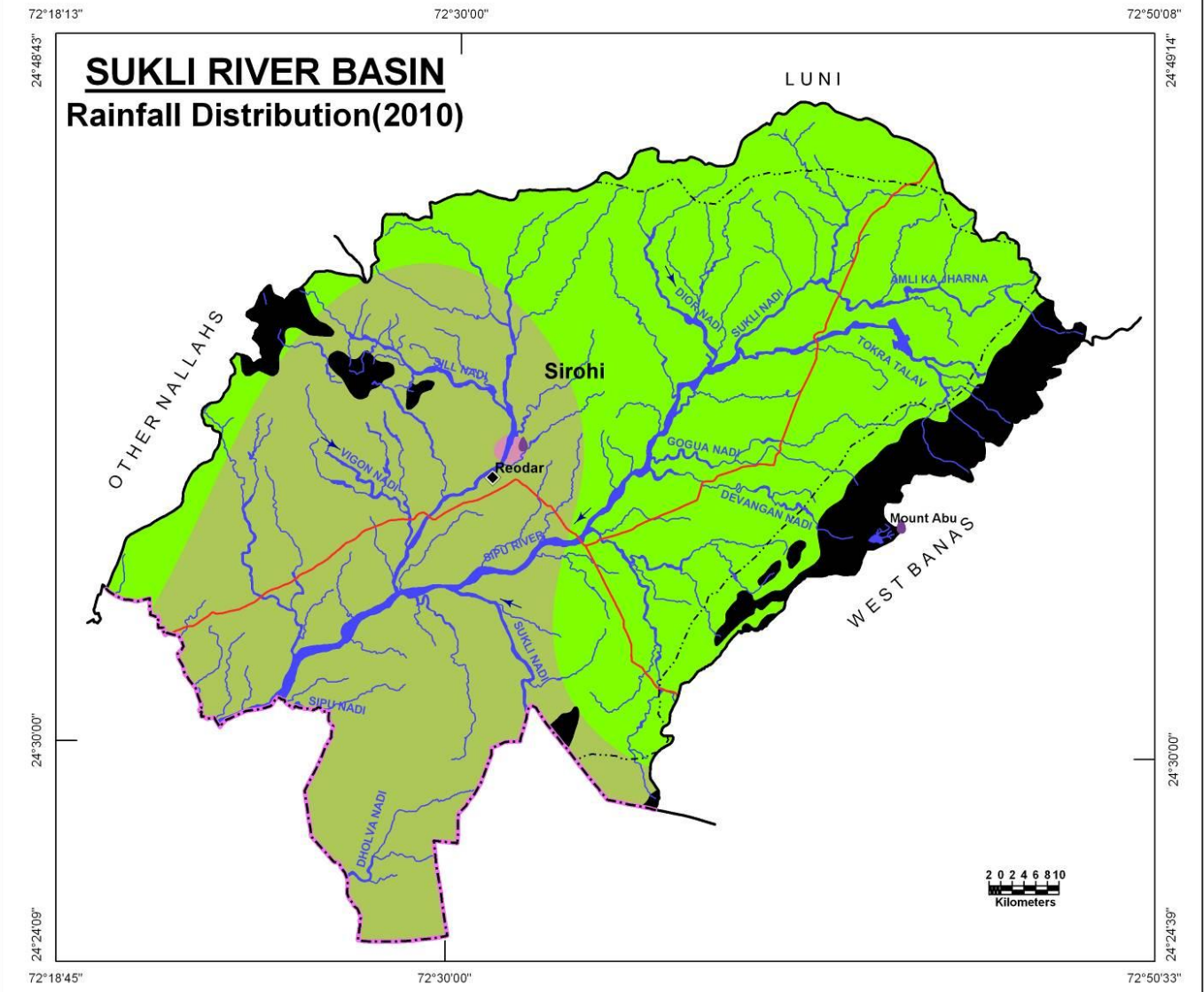
Ponds / Reservoirs

Elevation(m amsl):



Source : SRTM DEM

PLATE - III



LEGEND

Admin Boundary:

Block Headquarter

Raingauge Station

State Boundary

Block Boundary

River Basin Boundary

Roads:

State Highway

Water Bodies:

River / Streams

Ponds / Reservoirs

Hills

Isohyets(mm):



GEOLOGY

SUKLI RIVER BASIN

The Sukli River Basin is covered mainly by rocks belonging to the Pre-Cambrian rocks (Delhi and Aravali Super-Group) to Aeolian and Fluvial deposits of Recent to Sub-Recent age. Very limited area, i.e., valleys intervening hills has deposits of eolian and fluvial origin of Recent to Sub-Recent age.

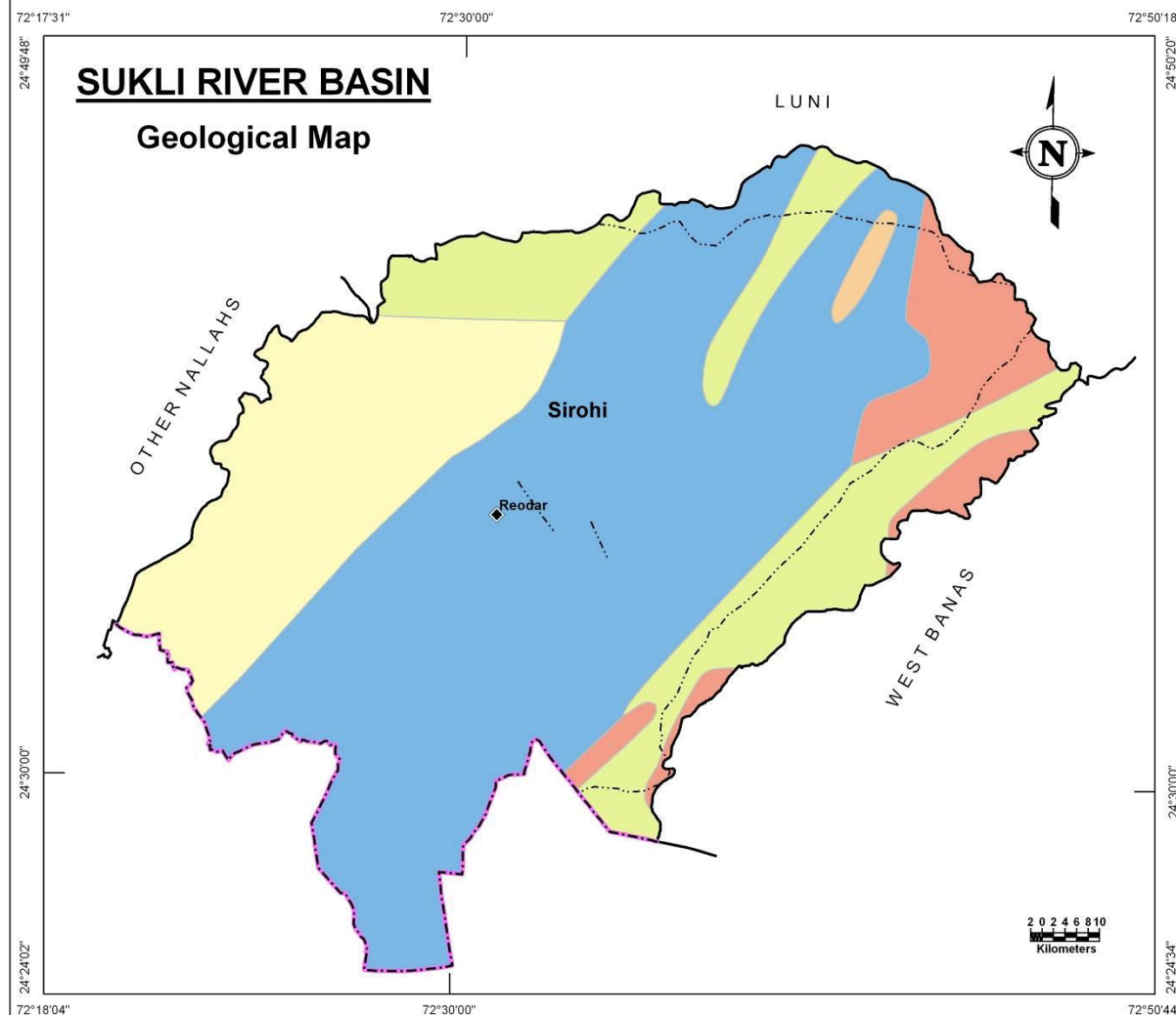
Age	Super-Group	Group/ Formation	Rock Types
Recent to Sub recent		Alluvium	Stream laid deposits, sand and gravel, mixed with clay Silt and kankar
-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----			
Lower to Upper Precambrian	Post Delhi	Malani	Rhyolite Dolerites, Basalt Jalore & Erinpura Granite
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----		
	Delhi Super-Group	Ajabgarh Series	Phyllite, Schist, Gneiss, Marble, Amphibolites, Calc Silitates, quartzite, mica schist etc.
-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----			
Lower Precambrian	Aravali		Phyllite, crystalline, Limestone, quartzite and Conglomerates with tuffs and lavas.

GEOMORPHOLOGY

Origin	Landform Unit	Description
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



PLATE - IV



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- Block Boundary

River Basin Boundary

Structural Features:

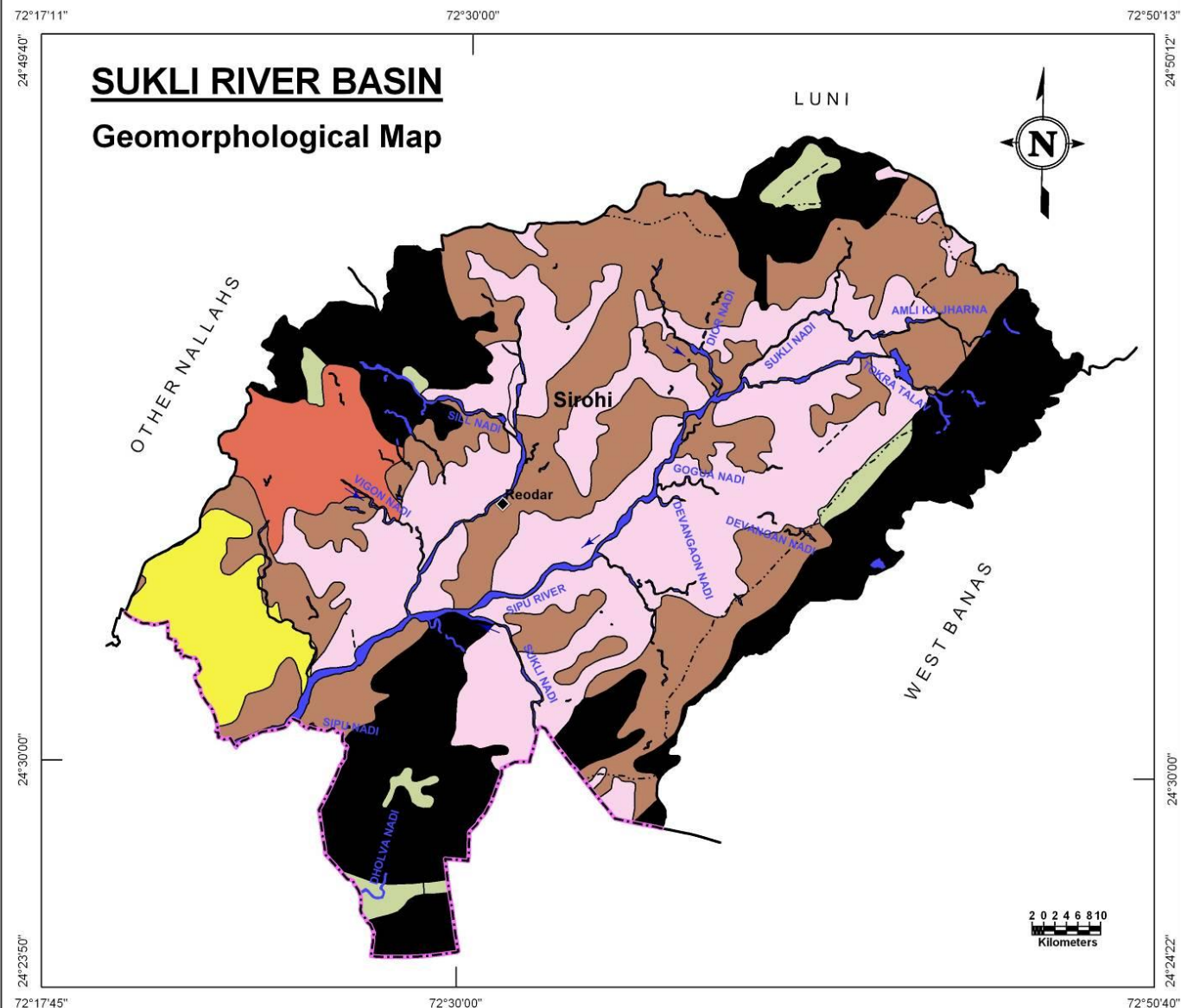
- Fault

Geology:

- Alluvium and Wind blown sand
- Sirohi Group
- Malani Plutonic
- Volcanic Suite
- Erinpura Granite & Gneiss

Source : District Resource Map of Rajasthan - GSI

PLATE - V



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- Block Boundary

River Basin Boundary

Water Bodies:

- River/Ponds / Reservoirs

Structural Features:

- Fault/Fractures/Lineament

Hills:

- Structural/Denudational/
Linear Ridge

Fluvial Origin:

- Alluvial Plain
- Valley Fill
- Ravine

Denudational Origin:

- Pediment
- Buried Pediment

Source : Ground Water Atlas of Rajasthan - SRSAC & GWD, Rajasthan

AQUIFERS

Weathered and fractured hardrocks seem to constitute most of the aquifer material in this basin. The phyllites occupy the central part of the basin over which the Sukli river seems to flow for its most part, accounting for more than 55% of the basin area. Granite aquifers flank the phyllite aquifers on either side towards southeast and northwest contributing to 31% of the basin's aquifer area. The younger alluvium aquifer forms a small part (less than 8%) in the southwestern part of the basin.

SUKLI RIVER BASIN

Aquifer in Potential Zone	Area (sq km)	% of Basin Area	Description of the unit/Occurrence
Younger Alluvium	75.6	7.6	It is largely constituted of Aeolian and Fluvial sand, silt, clay, gravel and pebbles in varying proportions.
Phyllite	550.1	55.1	These include meta sediments and represented by carbonaceous phyllite.
Granite	308.4	30.8	Light grey to pink colour, medium to coarse grained, and characteristically have porphyritic texture.
Non Potential Zone	65.3	6.5	Hills
Total	999.4	100.0	

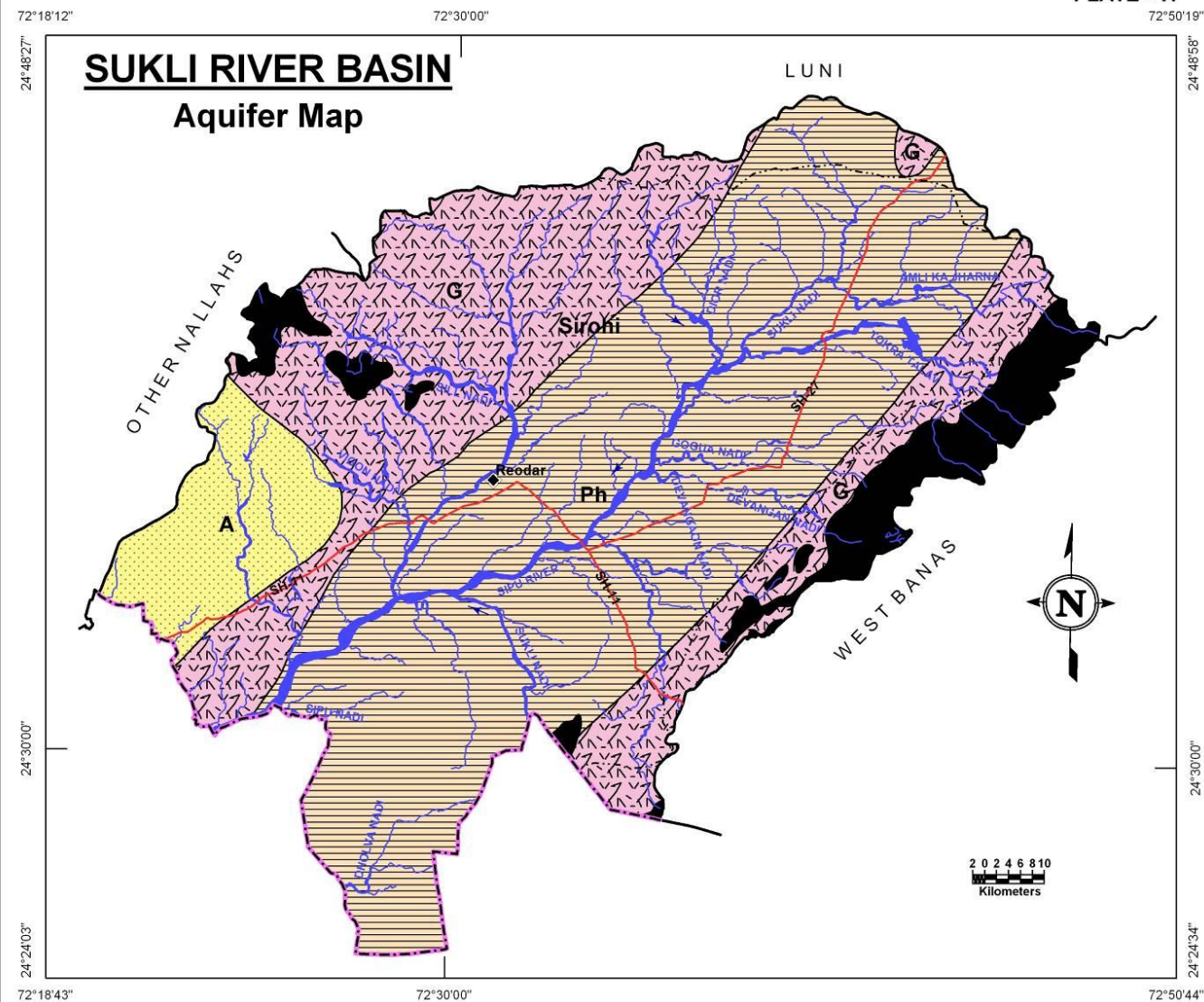
LOCATION OF GROUND WATER MONITORING WELLS

The basin has a fairly well distributed network of ground water monitoring stations (49) in the basin all owned by RGWD as this basin does not have any monitoring well belonging to CGWB network. After benchmarking of monitoring network, one additional well has been recommended to be added to network to effectively monitor ground water quality in the basin. The well network seems to have sufficient number of wells for water level monitoring.

District Name	CGWB	RGWD	Total	Recommended additional wells for optimization of monitoring network	
				Water Level	Water Quality
Sirohi	-	49	49	-	1



PLATE - VI



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- Block Boundary
- River Basin Boundary
- Roads:
- State Highway

Water Bodies:

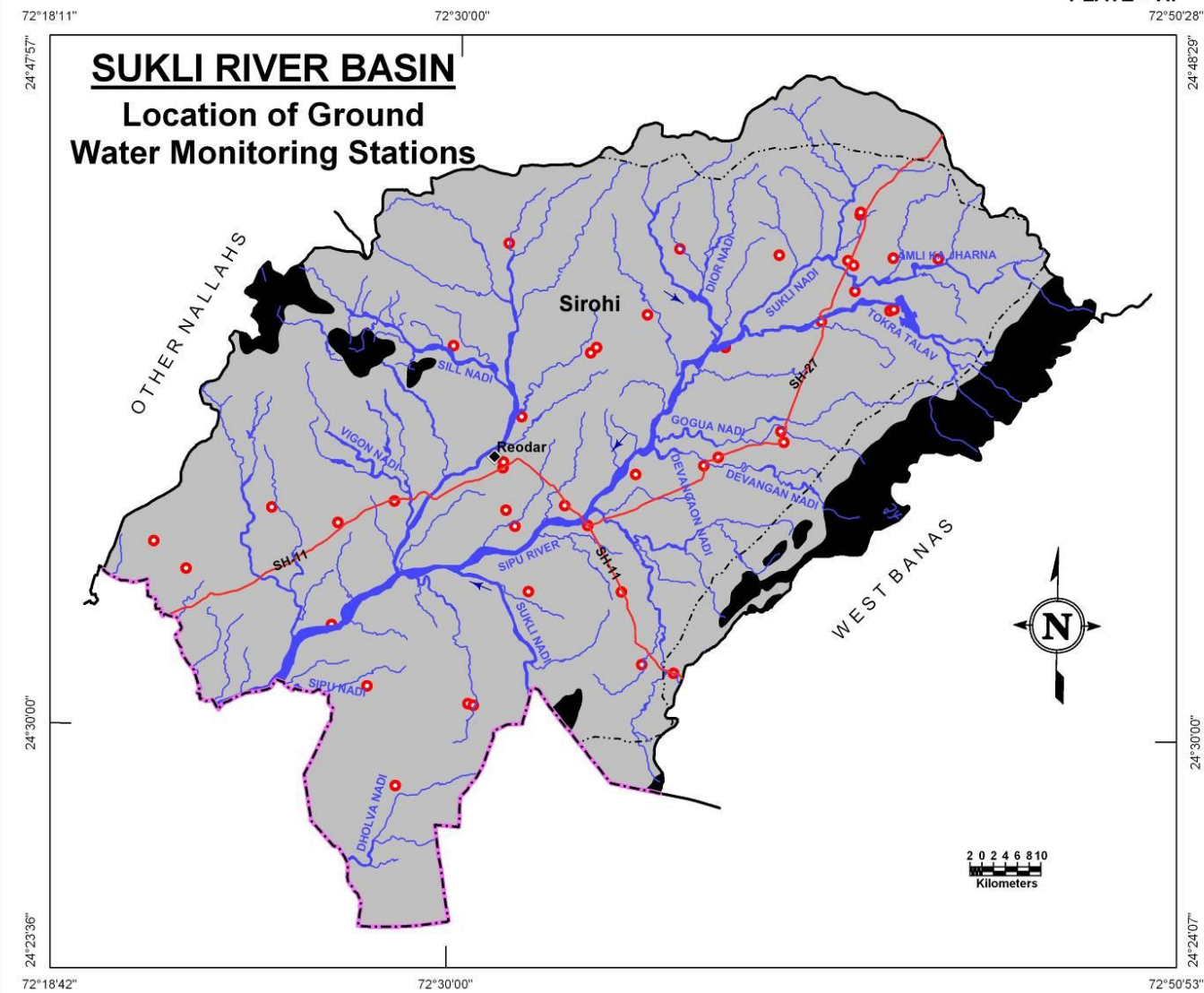
- River / Streams
- Ponds / Reservoirs
- Hills

Aquifer:

- Younger Alluvium
- Phyllite
- Granite

Source: Ground Water Potential Zone Map - GWD, Rajasthan

PLATE - VII



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- Block Boundary
- River Basin Boundary
- Roads:
- State Highway

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Ground Water Monitoring Stations:

- RGWD

LOCATION OF EXPLORATORY WELLS

In all there are 34 exploratory boreholes present in the basin drilled in the past by RGWD (33) and only one by CGWB that form basis for delineation of sub-surface aquifer distribution. Map (Plate – VIII) reveals that while there are good number of exploratory wells in the eastern part, there are very few or no wells in the western part of the basin.

District Name	CGWB	RGWD	Total
Sirohi	1	33	34

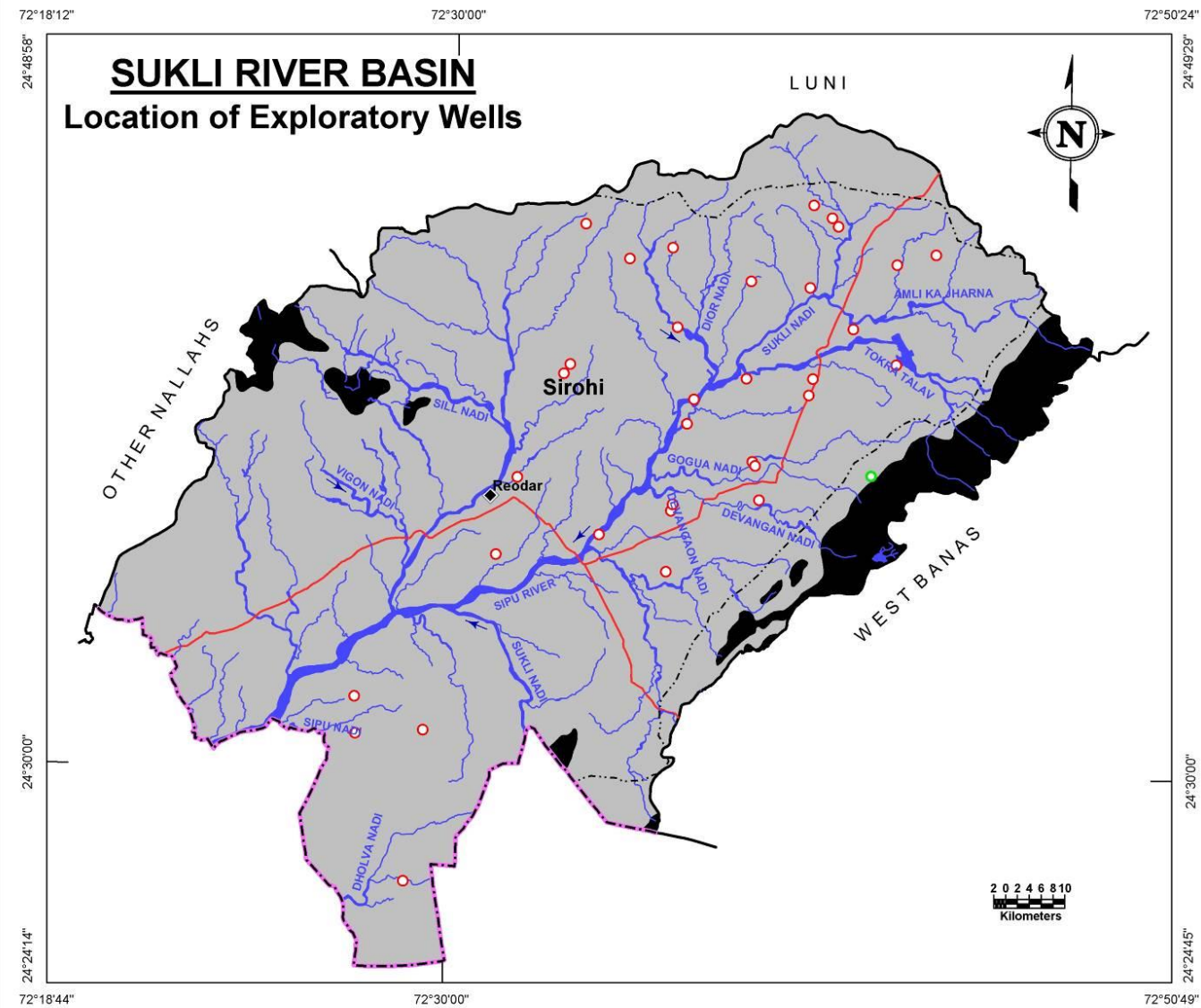
DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The depth to water level map is presented in Plate – IX. The general depth to water level in the basin ranges from 10 to 30 meters below ground level, as seen in most of the central parts and along the fringes which together account for more than 99% of the basin area. The small patch in the western most point of the basin adjacent to Other Nallahs Basin has recorded a much deeper water level (in the 30m bgl to 40m bgl).

Depth to water level (m bgl) Pre Monsoon – 2010	District wise area Coverage (sq km)*
	Sirohi
10 – 20	237.1
20 – 30	689.5
30 – 40	7.5
Total	934.1

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.

PLATE - VIII



LEGEND

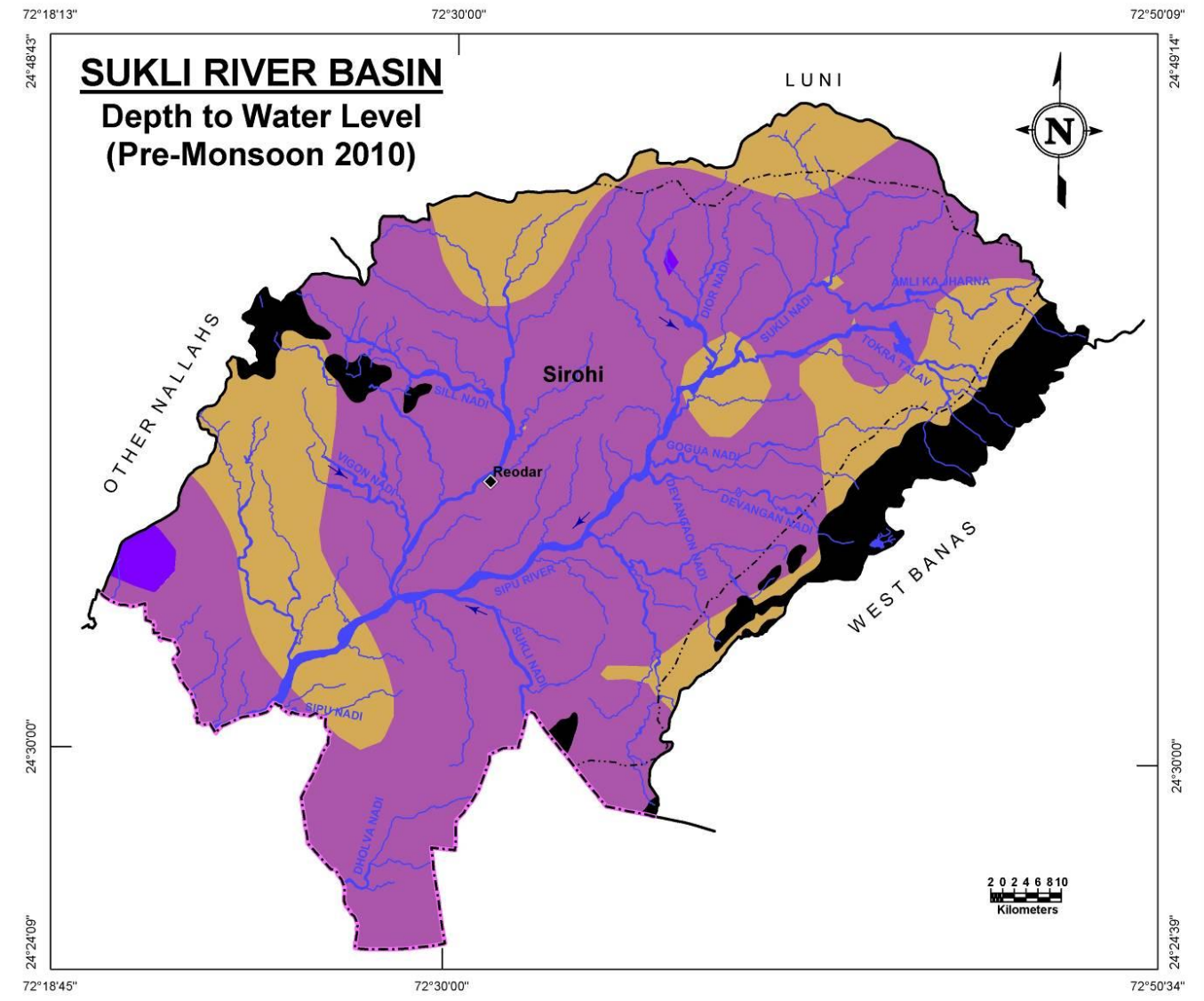
Admin Boundary:

- Block Headquarter
- State Boundary
- Block Boundary
- River Basin Boundary
- Roads:**
- State Highway
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

Exploratory Wells:

- CGWB
- RGWD

PLATE - IX



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- Block Boundary
- River Basin Boundary
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

Depth to Water Level (m bgl):

- 10 - 20
- 20 - 30
- 30 - 40

WATER TABLE ELEVATION (PRE MONSOON 2010)

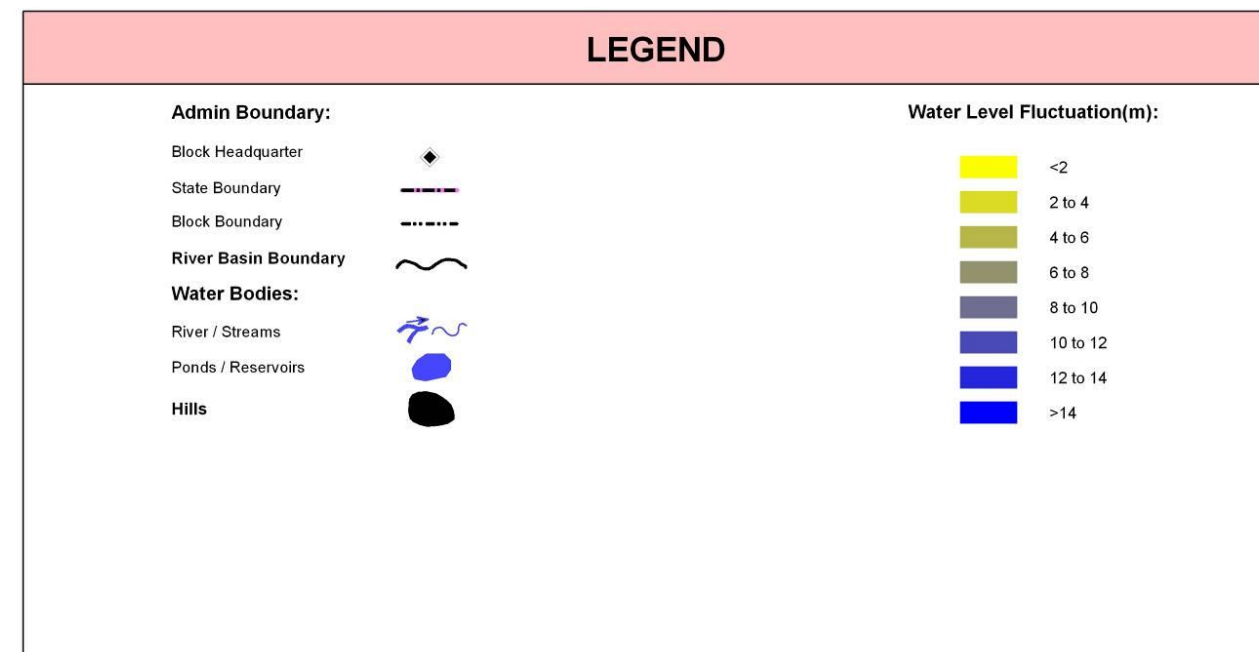
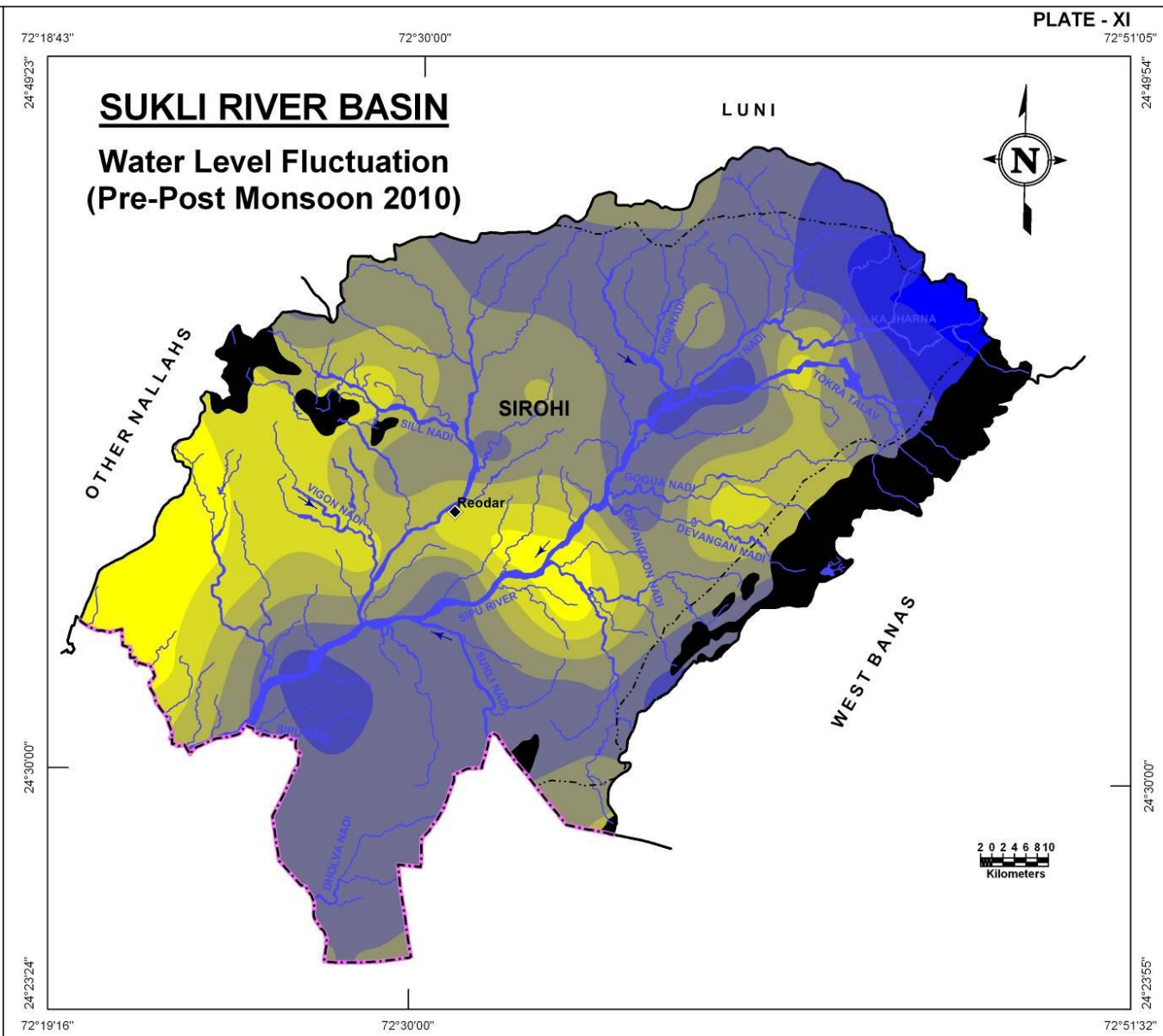
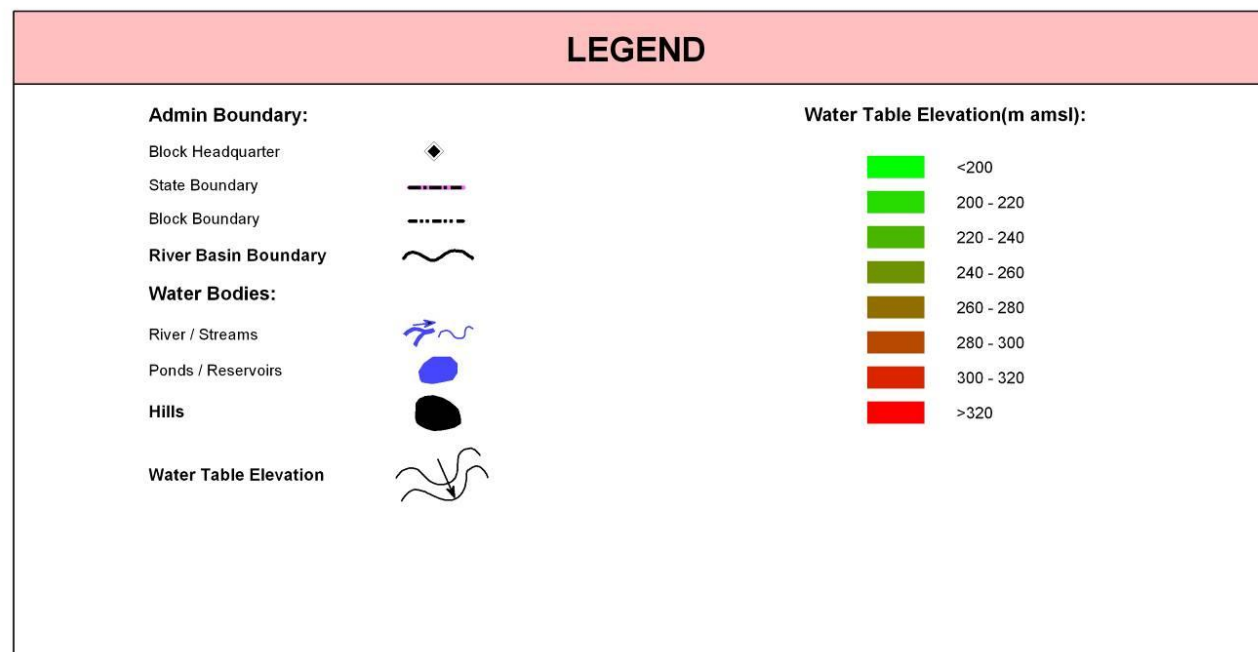
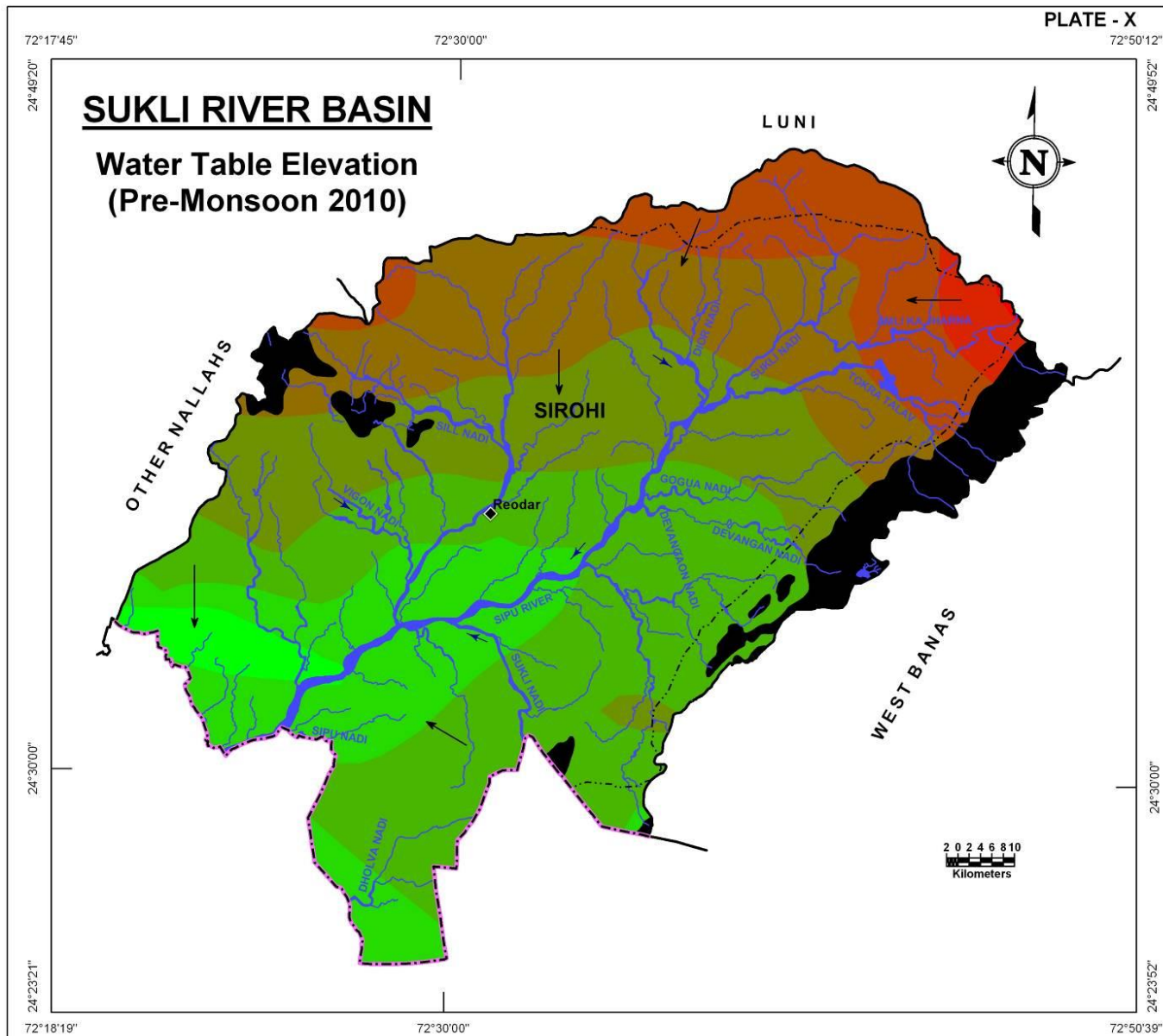
Water table elevation shows variation between about 320m amsl and 200m amsl. The water table contour map is presented in Plate – X. The highest water table is indicated in the northeastern most part of the basin and the lowest water table elevation is seen in the southwestern most part which suggests a general northeast to southwestwards flow of ground water in the basin. The arrows indicated in the map show the local flow directions which point towards southwest, south and southeast in the northern part all flowing towards the river channel whereas in the southern part the general ground water flow direction is east to west in the southeastern part and a north-south flow direction in the western part. Flow gradient is relatively steeper in the northern part as compared to southern part which has sluggish flow of ground water.

Water Table Elevation (m amsl) Pre Monsoon - 2010	District wise area coverage (sq km)
	Sirohi
< 200	24.5
200 - 220	149.3
220 - 240	283.4
240 - 260	178.9
260 - 280	177.6
280 - 300	103.7
300 - 320	16.2
> 320	0.5
Total	934.1

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

The Basin is primarily a hardrock aquifer terrain and the aquifers formed in alluvial area are very limited in distribution. The water level fluctuation map is presented in Plate – XI which indicates a large fluctuation range from less than 2m to more than 14m. The area with highest rise in ground water level is located in the northeastern part of the basin where the river originates and is also the area where maximum rainfall in the basin was recorded. There are two pockets, one in the western part where alluvial aquifer is present and the other in the centre of the basin where fluctuation has been in the range of less than 2m to 6m rise whereas the rest of the basin has shown a general rise in ground water level between 6m to 10m.

District Name	District wise area coverage (sq km) within fluctuation range (m)								Total Area (sq km)
	< 2	2 to 4	4 to 6	6 to 8	8 to 10	10 to 12	12 to 14	> 14	
Sirohi	56.4	104.2	137.6	220.1	324.0	59.8	21.5	10.5	934.1



ELECTRICAL CONDUCTIVITY DISTRIBUTION

SUKLI RIVER BASIN

The Electrical Conductivity (at 25°C) distribution map is presented in Plate XII. Perusal of the map reveals only yellow (EC < 2000 $\mu\text{S}/\text{cm}$) and green (EC 2000 – 4000 $\mu\text{S}/\text{cm}$) coloured regions suggesting that about 90% of the area has good quality of ground water suitable for domestic purposes and the rest i.e., about 10% area has moderately high EC that is still within tolerable limits for potability. There are no areas that have shown very high EC in ground water (EC >4000 $\mu\text{S}/\text{cm}$).

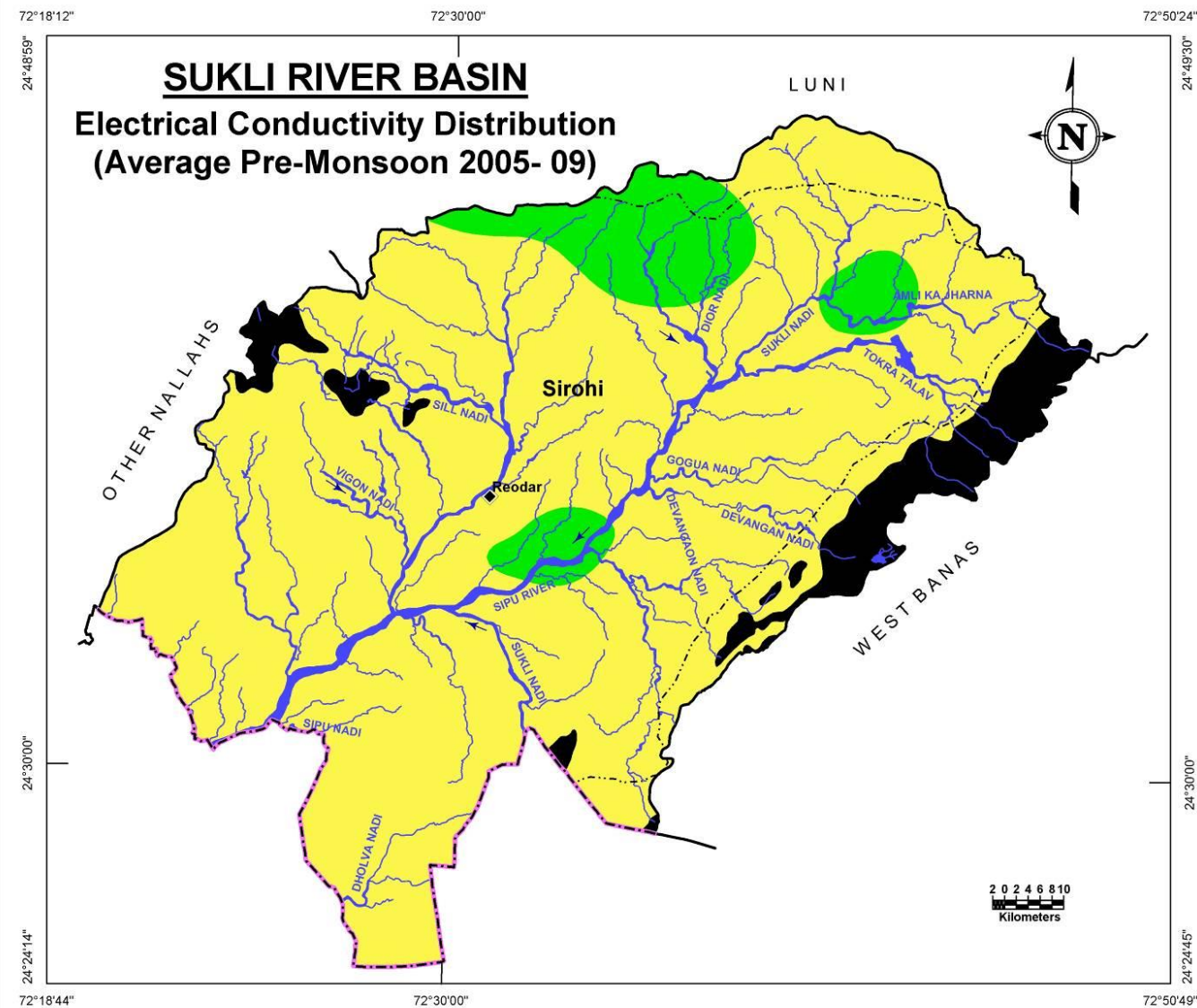
Electrical Conductivity Ranges ($\mu\text{S}/\text{cm}$ at 25°C) (Ave. of years 2005-09)	District wise area coverage (sq km)	
	Sirohi	
	Area	% age
< 2000	845.7	90.5
2000-4000	88.3	9.5
> 4000	0.1	-
Total	934.1	100.0

CHLORIDE DISTRIBUTION

Plate – XIII illustrates Chloride distribution map. Moderately high concentration (250 – 1000 mg/l) areas are seen in north, central and a small patch in the southern part mostly within the phyllite aquifers. Such areas occupy about 35% of the basin area. The remaining part of the basin has shown the presence of low concentration of chloride in ground water (<250 mg/l) rendering it suitable for all purposes. There is also seen a very small (<1 sq km) patch of red colour marking the presence of high chloride concentration in ground water in the northern part of the basin.

Chloride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)	
	Sirohi	
	Area	% age
< 250	611.4	65.5
250 - 1000	322.1	34.5
> 1000	0.6	0.0
Total	934.1	100.0

PLATE - XII



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

Block Boundary

River Basin Boundary

Water Bodies:

River / Streams

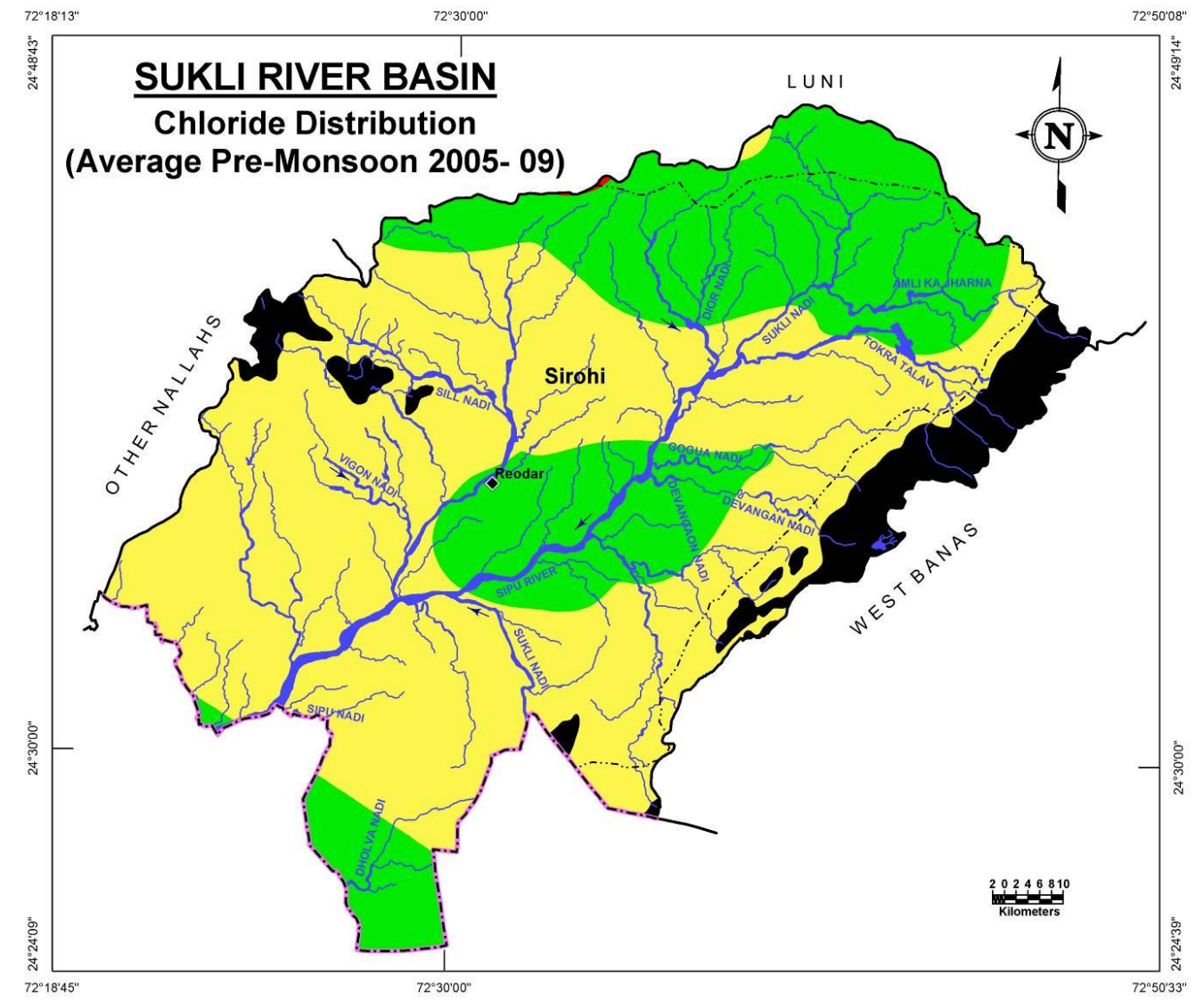
Ponds / Reservoirs

Hills

Electrical Conductivity ($\mu\text{S}/\text{cm}$ at 25°C):



PLATE - XIII



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

Block Boundary

River Basin Boundary

Water Bodies:

River / Streams

Ponds / Reservoirs

Hills

Chloride Concentration (mg/l):



FLUORIDE DISTRIBUTION

The Fluoride concentration map (Plate – XIV) displays a number of scattered patches of high fluoride concentration (>3 mg/l) which is surrounded but an even larger area having 1.5 – 3.0 mg/l of fluoride in ground water. Together these two areas combined (i.e., > 1.5 mg/l), occupy close to 84% of the basin rendering the ground water of limited use for domestic purposes from fluoride concentration point of view. Interestingly these high Fluoride (>3.0 mg/l) areas correspond to the granitic aquifers as can be seen from Plate – VI. Relatively good quality of ground water (i.e., having <1.5 mg/l of Fluoride concentration) is only present in areas that have aquifers formed in either Phyllite or in alluvium. The southward bulge of propagation of fluoride is possibly due to the southwards flow of ground water in that region towards Sukli River.

Fluoride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)	
	Sirohi	
	Area	% age
< 1.5	305.1	32.6
1.5-3.0	476.8	51.1
> 3.0	152.2	16.3
Total	934.1	100.0

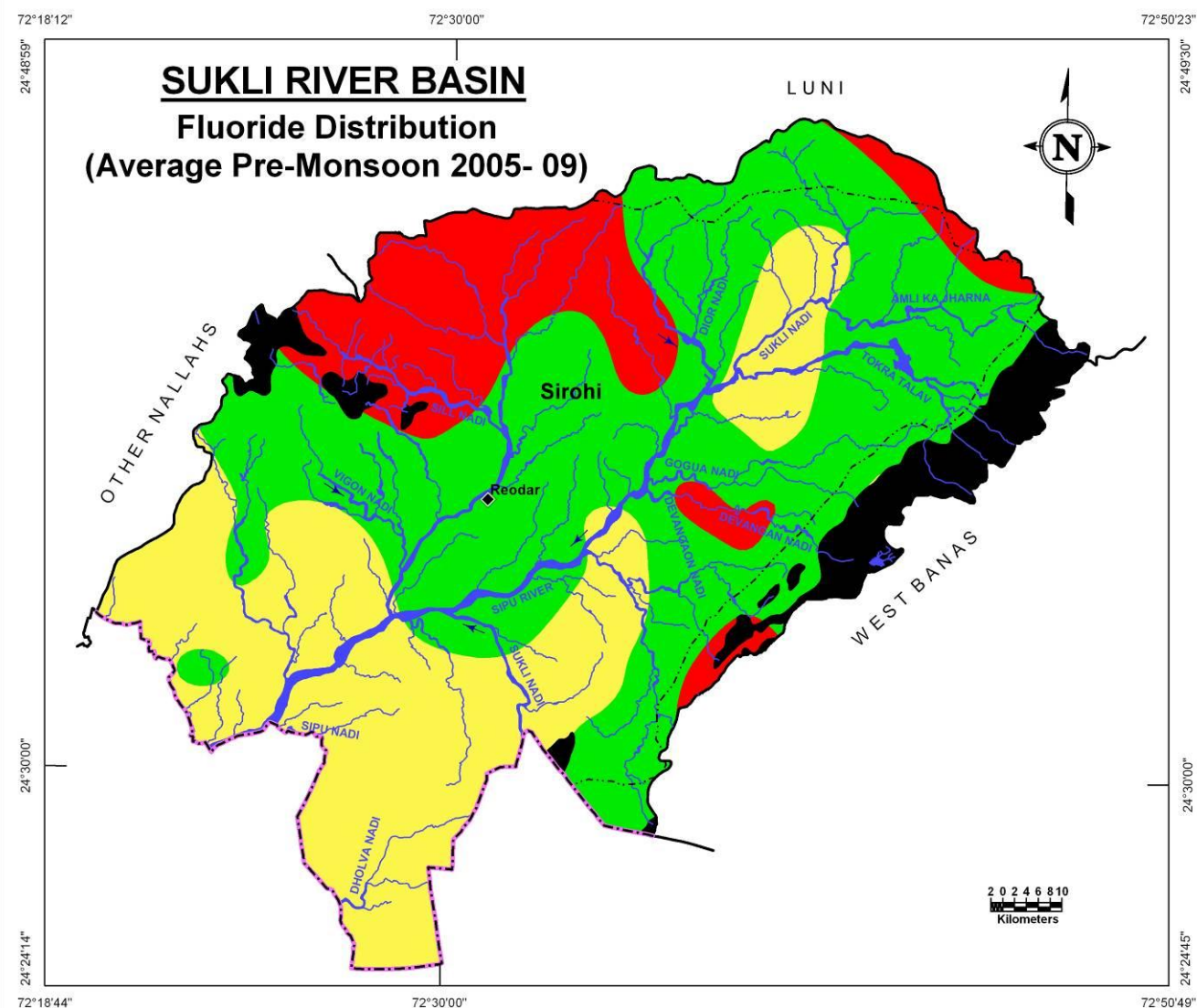
NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of Nitrate in ground water. The Nitrate concentration is also high in this basin as seen by the red (>100mg/l) and green (50 – 100 mg/l) coloured regions occupying about 83% basin area. The patch seems to occupy large area in the north and propagate along the ground water flow direction towards south affecting major part of the area to fall into moderately high (50 – 100 mg/l). That leaves the ground water of only about 17% of the basin area suitable for agriculture purposes.

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)	
	Sirohi	
	Area	% age
< 50	483.2	51.8
50-100	295.5	31.6
> 100	155.4	16.6
Total	934.1	100.0



PLATE - XIV



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

District Boundary

Block Boundary

River Basin Boundary

Water Bodies:

River / Streams

Ponds / Reservoirs

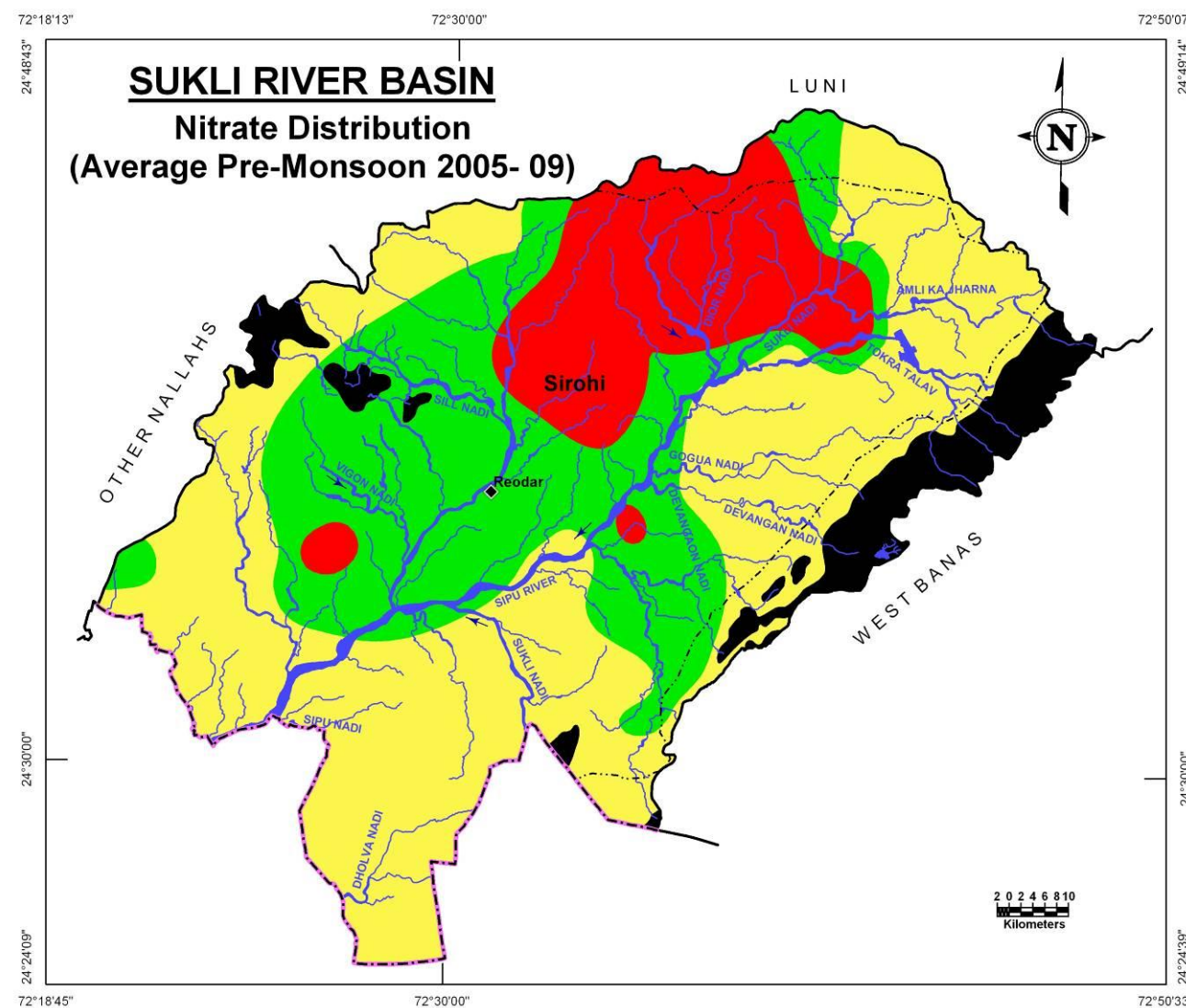
Hills

Fluoride Concentration (mg/l):



SUKLI RIVER BASIN

Nitrate Distribution
(Average Pre-Monsoon 2005- 09)



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

District Boundary

Block Boundary

River Basin Boundary

Water Bodies:

River / Streams

Ponds / Reservoirs

Hills

Nitrate Concentration (mg/l):



DEPTH TO BEDROCK

The hilly areas expose the bedrock and gradually into the plains, the thick alluvial pile conceals the bedrock under sand, clay and mix. From an aquifer perspective, the beginning of massive bedrock is taken to mark the start of bedrock and thus the weathered and fractured part of bedrock and alluvial cover constitutes the material above the bedrock. Plate – XVI represents depth to bedrock in meters below ground level (bgl). There are two isolated patches of depth to bedrock more than 40m bgl in the western border of the basin and one patch of less than 20m bgl in the eastern part adjacent to hills. These areas together add up to 4% of the basin area whereas the remaining 96% of the basin area has depth to bedrock ranging between 20 – 40m bgl.

Depth to Bedrock (m bgl)	District wise area coverage (sq km)
	Sirohi
< 20	16.4
20-40	896.5
> 40	21.2
Total	934.1

UNCONFINED AQUIFER

Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas. The same is presented in Plate – XVII.

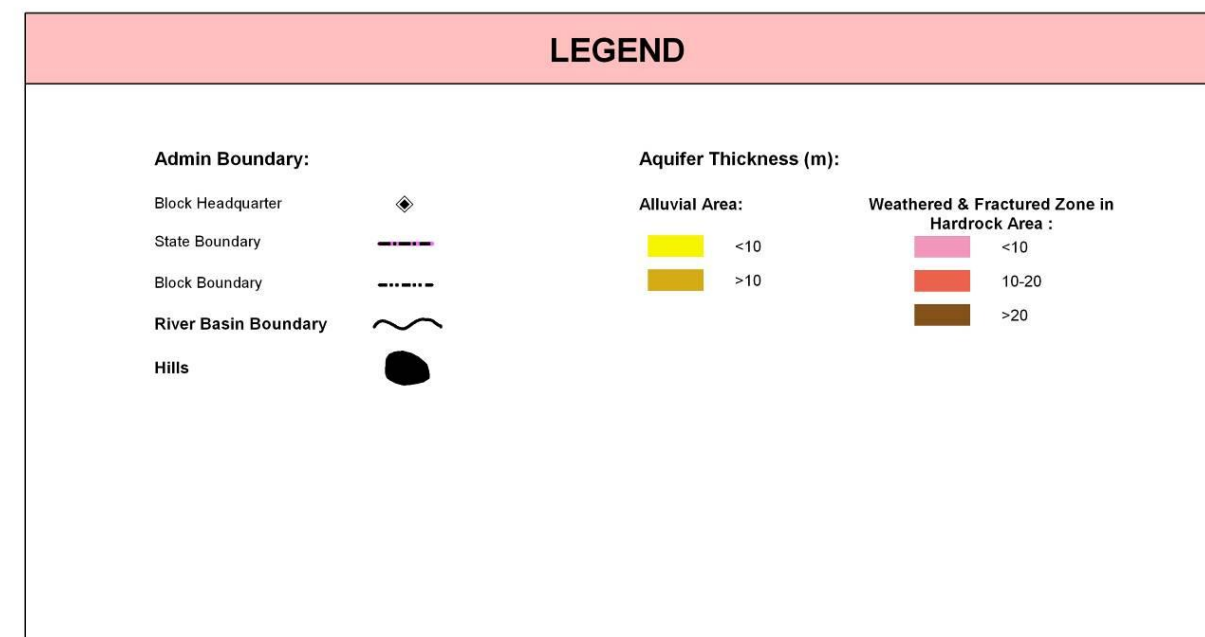
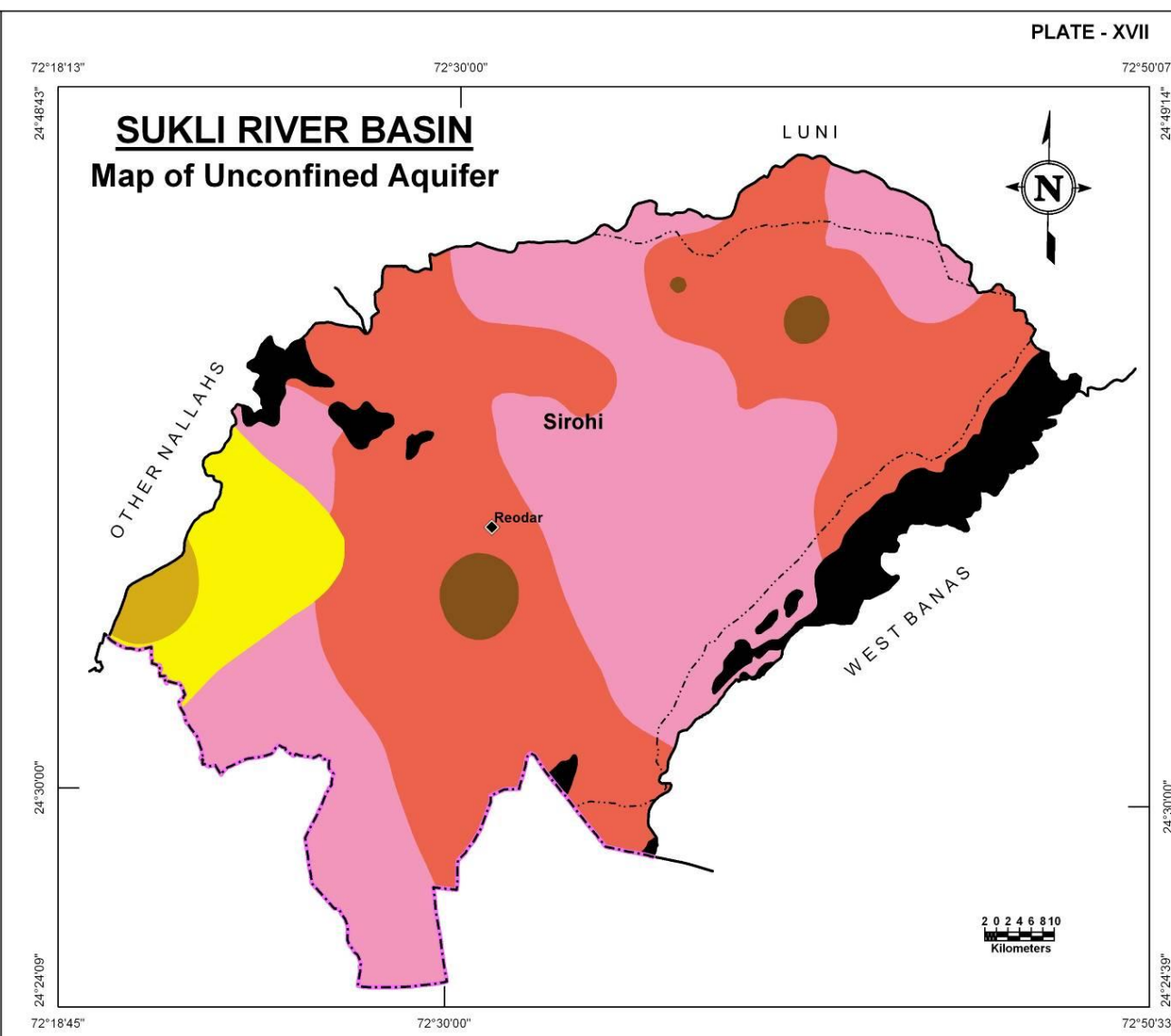
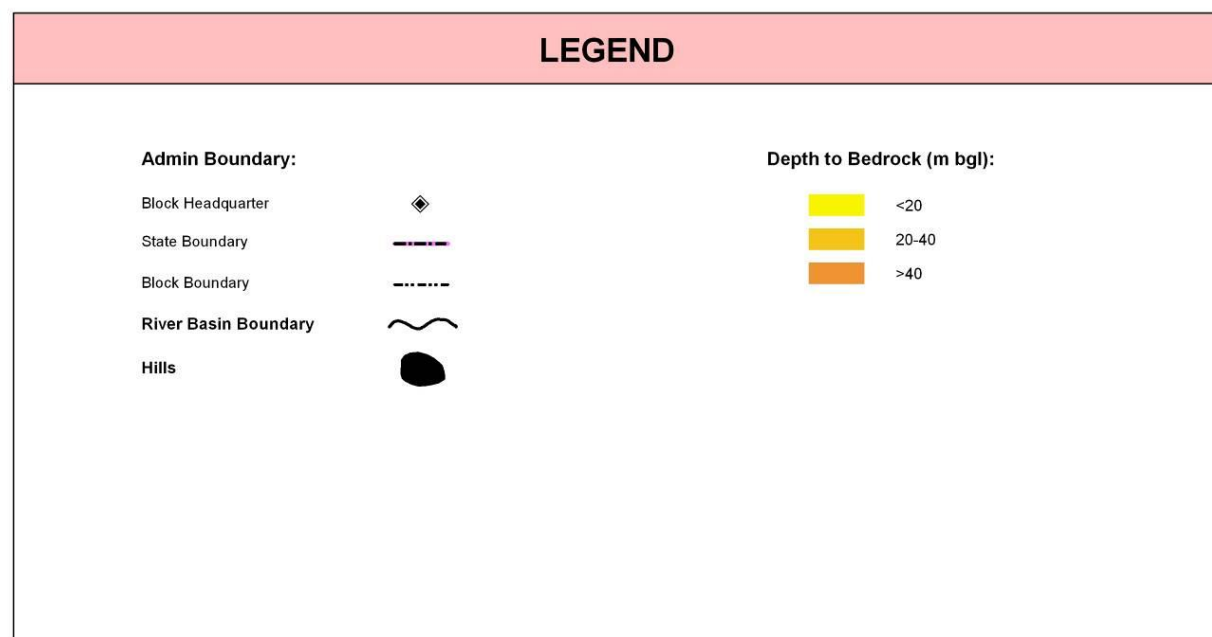
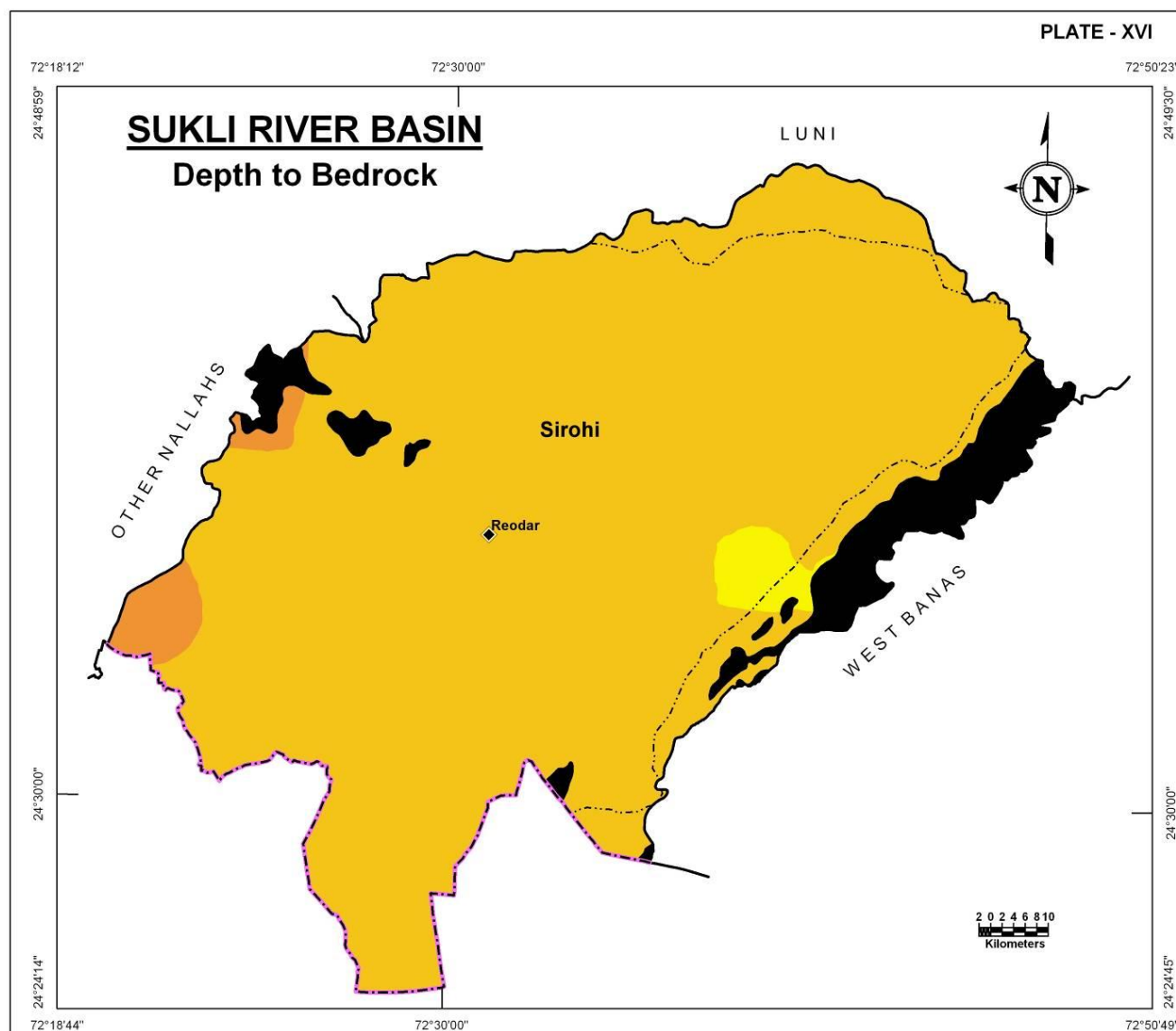
The alluvial aquifer is present in the southwestern part of the basin and is predominantly of eolian or fluvial origin represented by sand, clay and gravel. The thickness of unconfined aquifer in alluvial areas is often less than 10m (82% of alluvial part of the basin) and only about 17% of the alluvial area has slightly more thickness than 10m. In hardrock areas also the thickness of unconfined aquifer (weathered and fractured hard rock) is mostly less than 20m (almost 98% of the hardrock aquifer area) and very small patches in the centre and north have slightly more thickness than 20m.

Alluvial areas

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)
	Sirohi
< 10	62.9
> 10	12.7
Total	75.60

Hardrock areas:

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)
	Sirohi
<10	400.0
10-20	440.1
> 20	17.7
Total	857.80

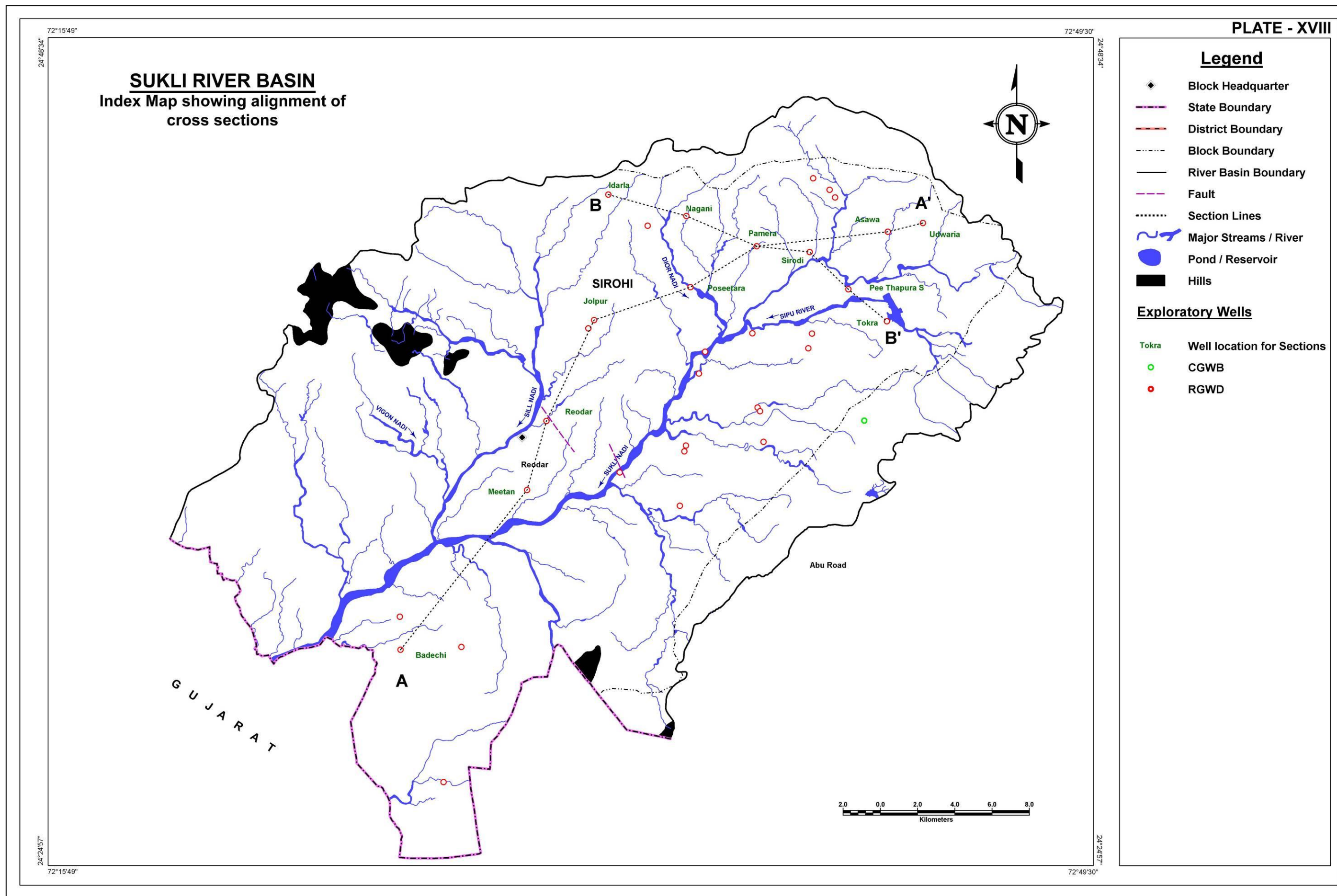


CROSS SECTIONS

SUKLI RIVER BASIN

Several hydrogeologic cross sections have been drawn to better decipher the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their impact on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XX. The broad alignment of the sections is as given below:

Name of Section Line	Orientation
Section AA'	SW – NE
Section BB'	W – E



CROSS SECTIONS

SUKLI RIVER BASIN

Section A-A':

The section A-A' (Plate – XIX) trends in SW-NE direction, extending to a length of about 30km. The lithologs of 6 boreholes along with surrounding well information has been taken into account while preparing the section. The section depicts a predominantly hardrock presence in the section. Schist is present from southwest to northeast in all the wells however, a tentative fault seems to have brought the granites in juxtaposition against schist. Small lens of dolerite is present in the southern part of the section but is of limited spatial distribution. Alluvium represented by sand is seen in the middle of the section but lies above the ground water level thus of not much significance from ground water storage perspective. Topography is higher in the northeast and lower in the southwest.

The water table elevation varies from 200 m amsl to 280 m amsl following the surface topography.

Section B-B':

The section B-B' (Plate – XX) has been selected to represent a NW-SE profile of the sub-surface covering a length of about 17.5km. The lithologs of 5 boreholes along with surrounding well information is taken into account while preparing the section. Granite is predominant in this section overlain by topsoil in most part of the section. In the southeastern part of the profile a Fault is present separating granite in the southeast from sand from phyllite to northwest. The alluvial sand however having about 20m in thickness, seem to have largely remained above water table and therefore may not be in a position to form aquifer in the area.

Water table elevation varies from 290 m amsl to 295 m amsl in this profile and water from both sides flows towards centre which also is the topographic low in the section.

PLATE - XIX

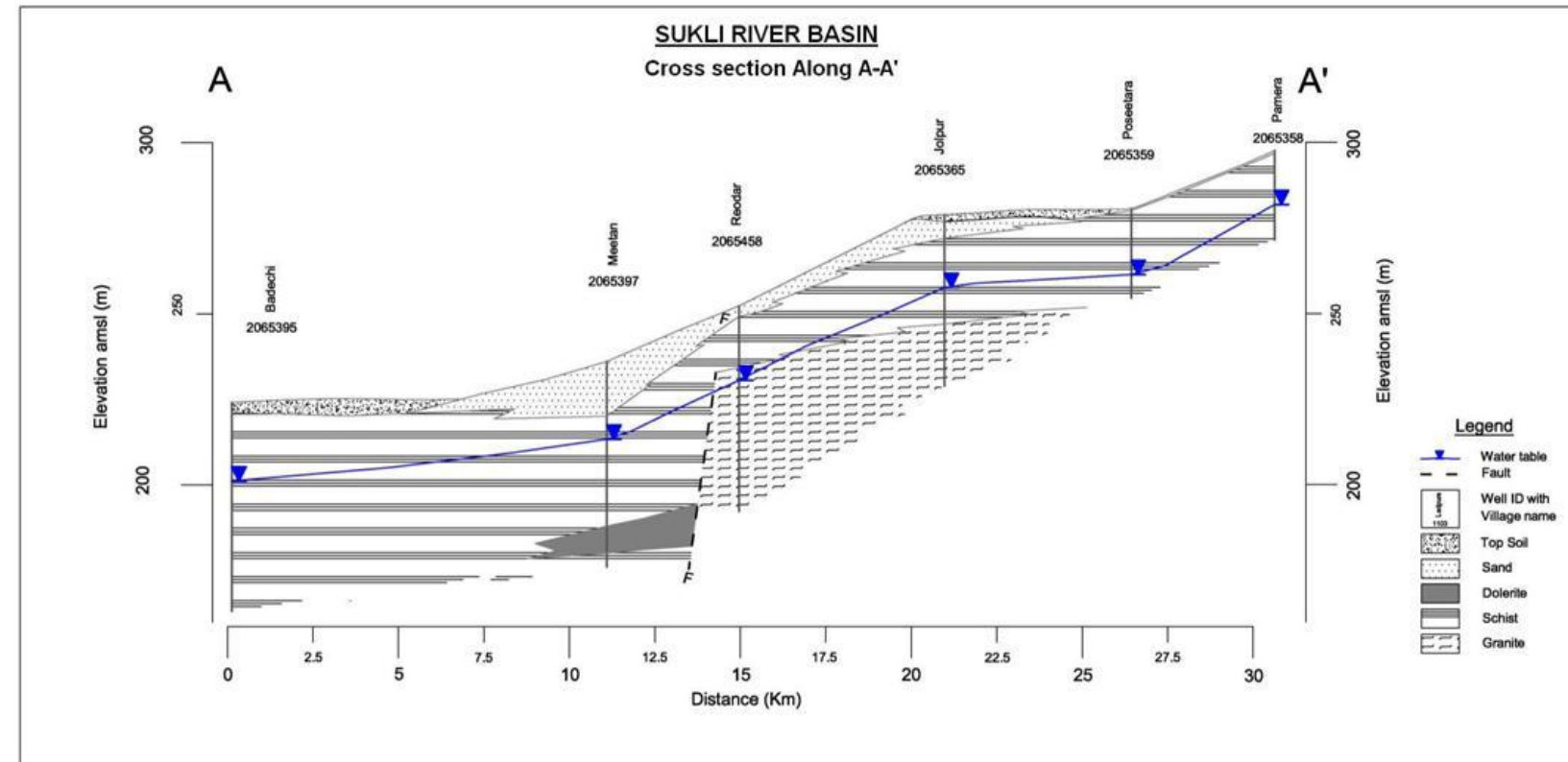
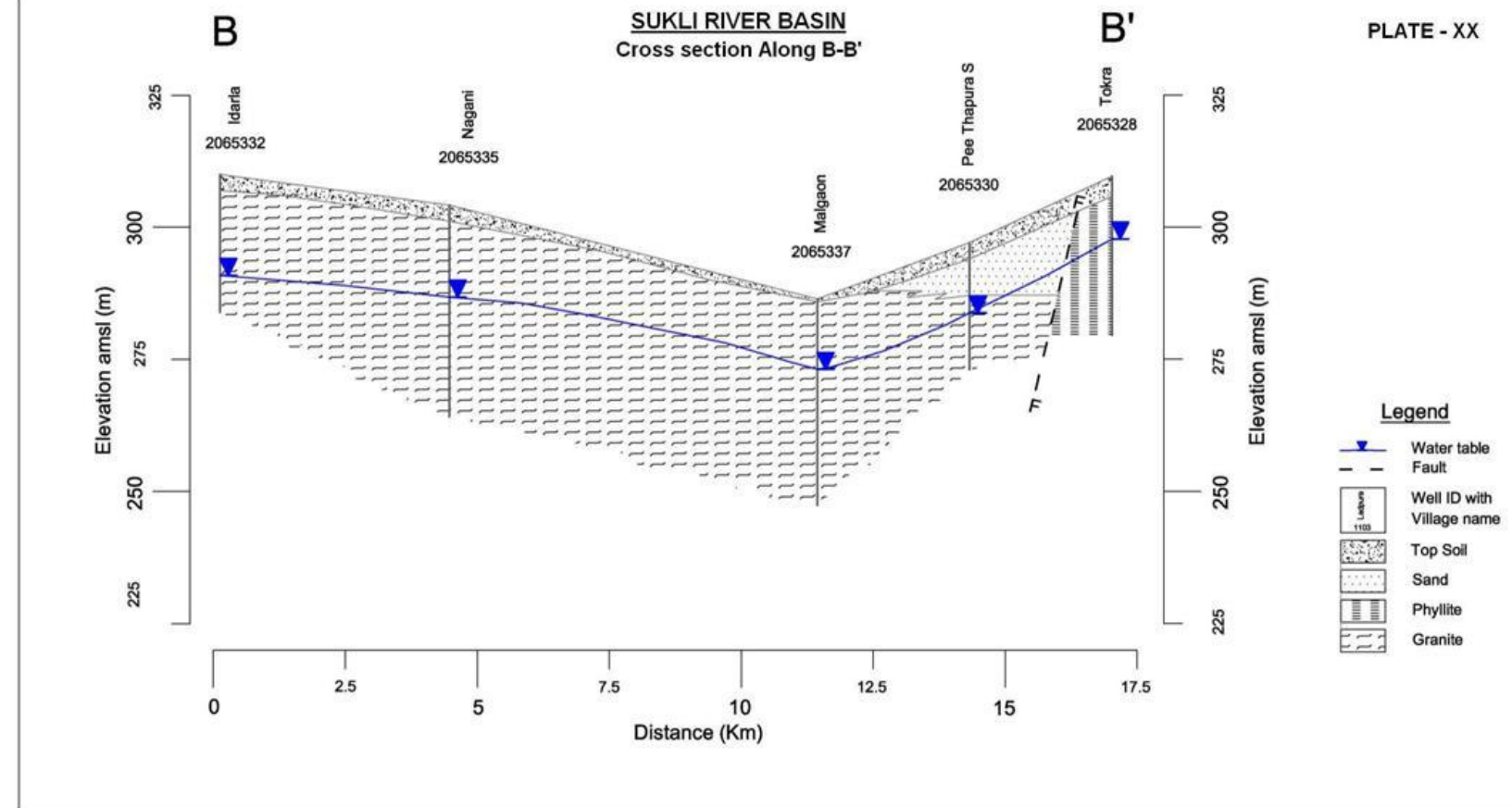


PLATE - XX



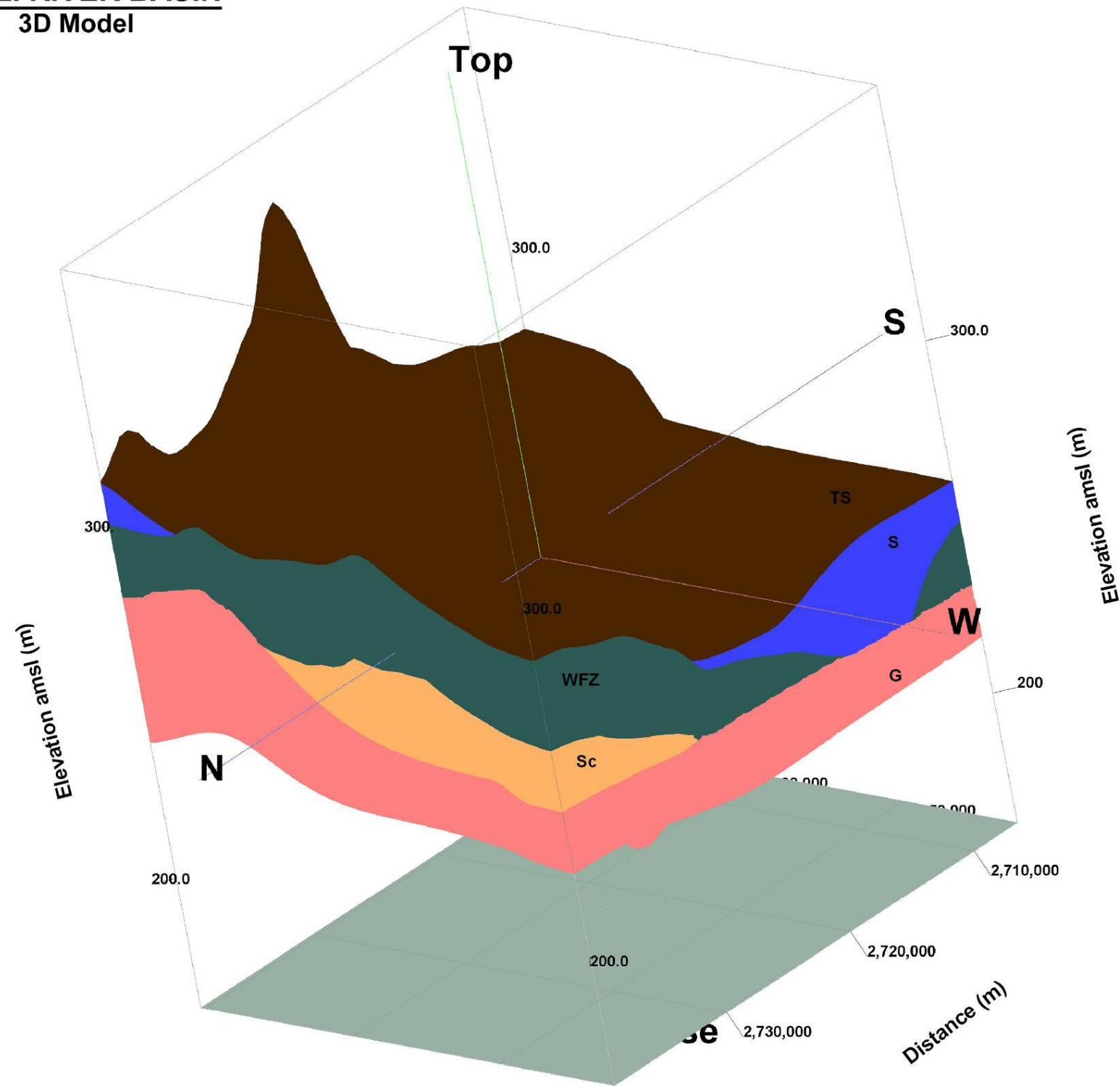
3D MODEL OF AQUIFERS

SUKLI RIVER BASIN

The continuous litho-stratigraphic model has been developed for the Sukli River Basin using the data of scattered exploratory boreholes as input. 3D model depicts the sub-surface aquifer disposition of litho-stratigraphic units forming aquifers, aquicludes and aquitards in the area. Plate –XXI presents 3D model depicting the various litho-stratigraphic units in the entire river basin. From this model it is apparent that beneath the top soil there is a persistent weathered and fractured bed rock acting as unconfined aquifers. Alluvium consisting mainly of sand, having limited occurrence in the western part of the area is acting as unconfined alluvial aquifer in the basin. Gneiss is underlying the weathered and fractured hard rock zone and alluvium of the area, behaving as an aquitard, is helping the overlying alluvial and hard rock aquifer to hold water. Schist, Phyllite and Dolerite overlay the Granites and when fractured, constitute good aquifers in the basin.

PLATE - XXI

SUKLI RIVER BASIN
3D Model



Legend

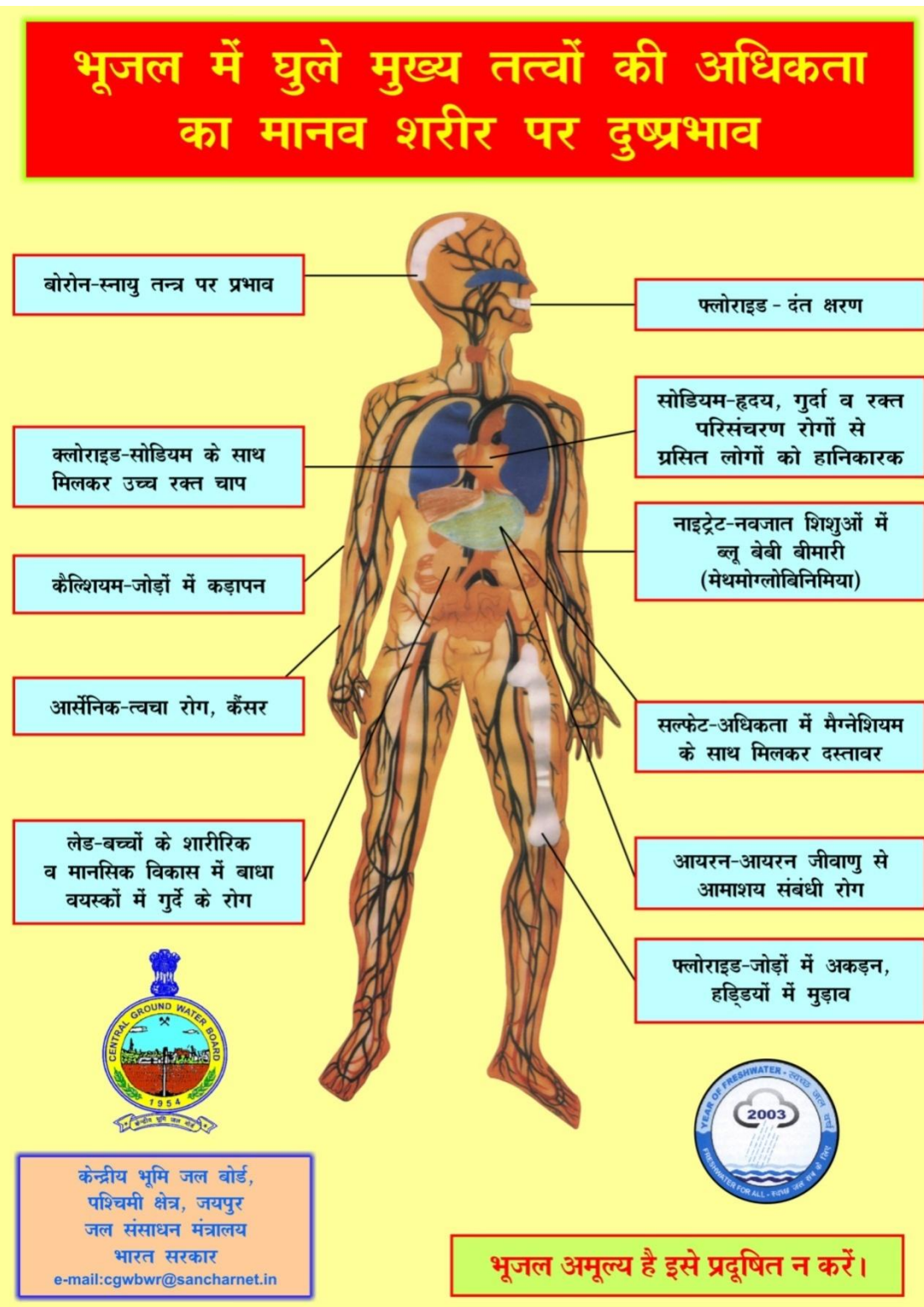
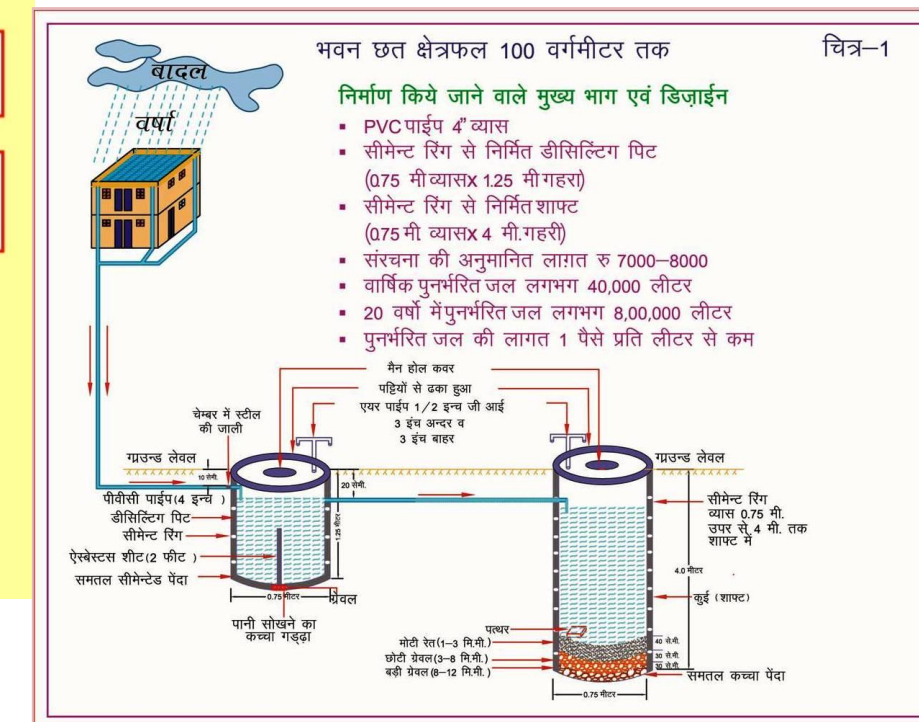
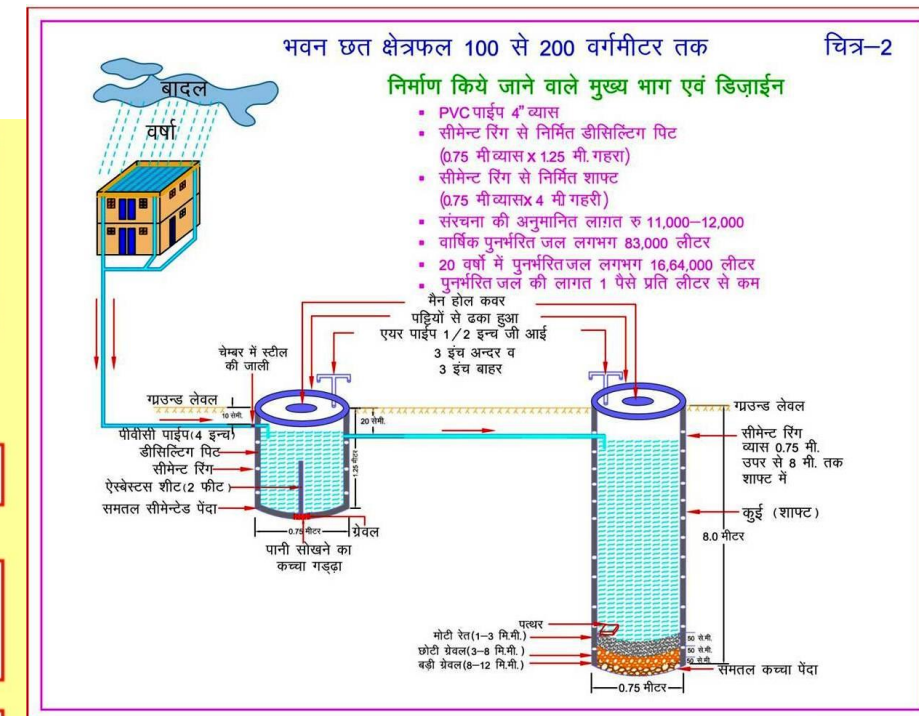
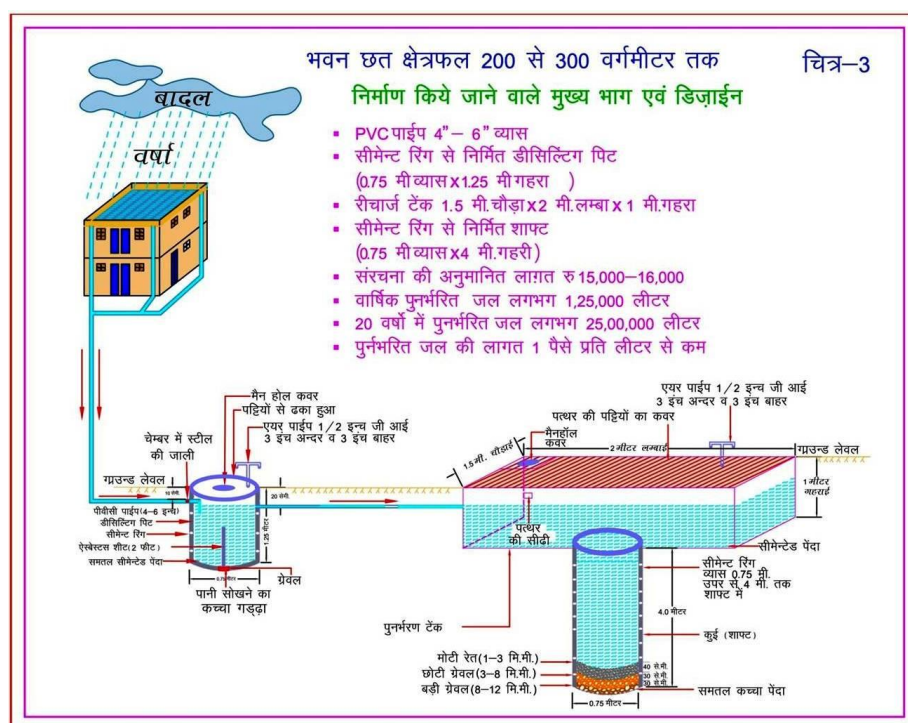
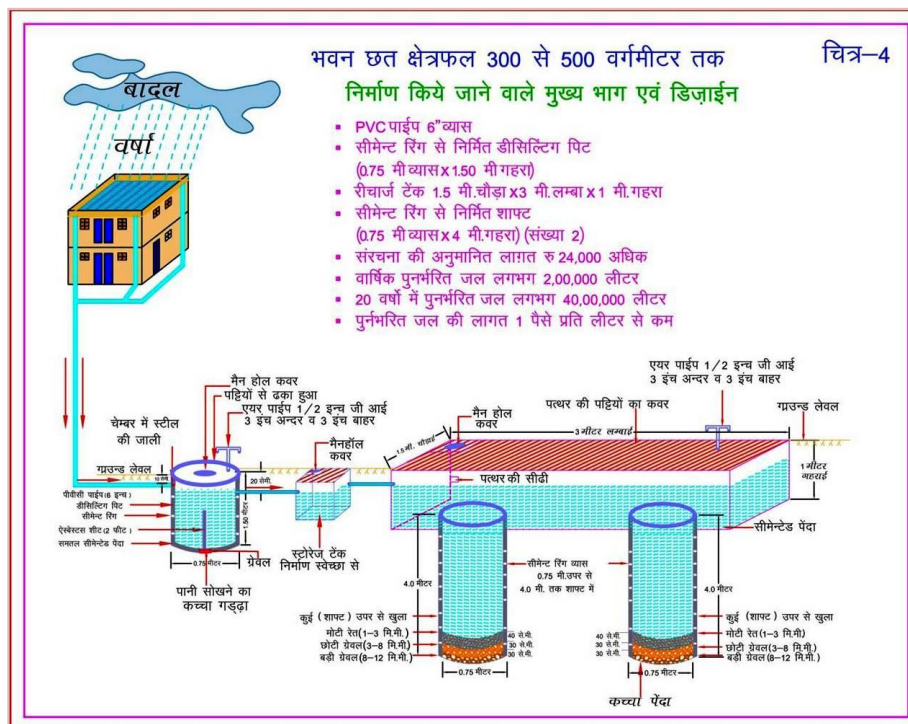
TS	Top Soil
S	Sand
WFZ	Weathered & Fractured Zone
D	Dolerite
Ph	Phyllite
Sc	Schist
G	Granite

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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